

经济高效注意力架构的 稀疏性、低秩性及位置冗余性

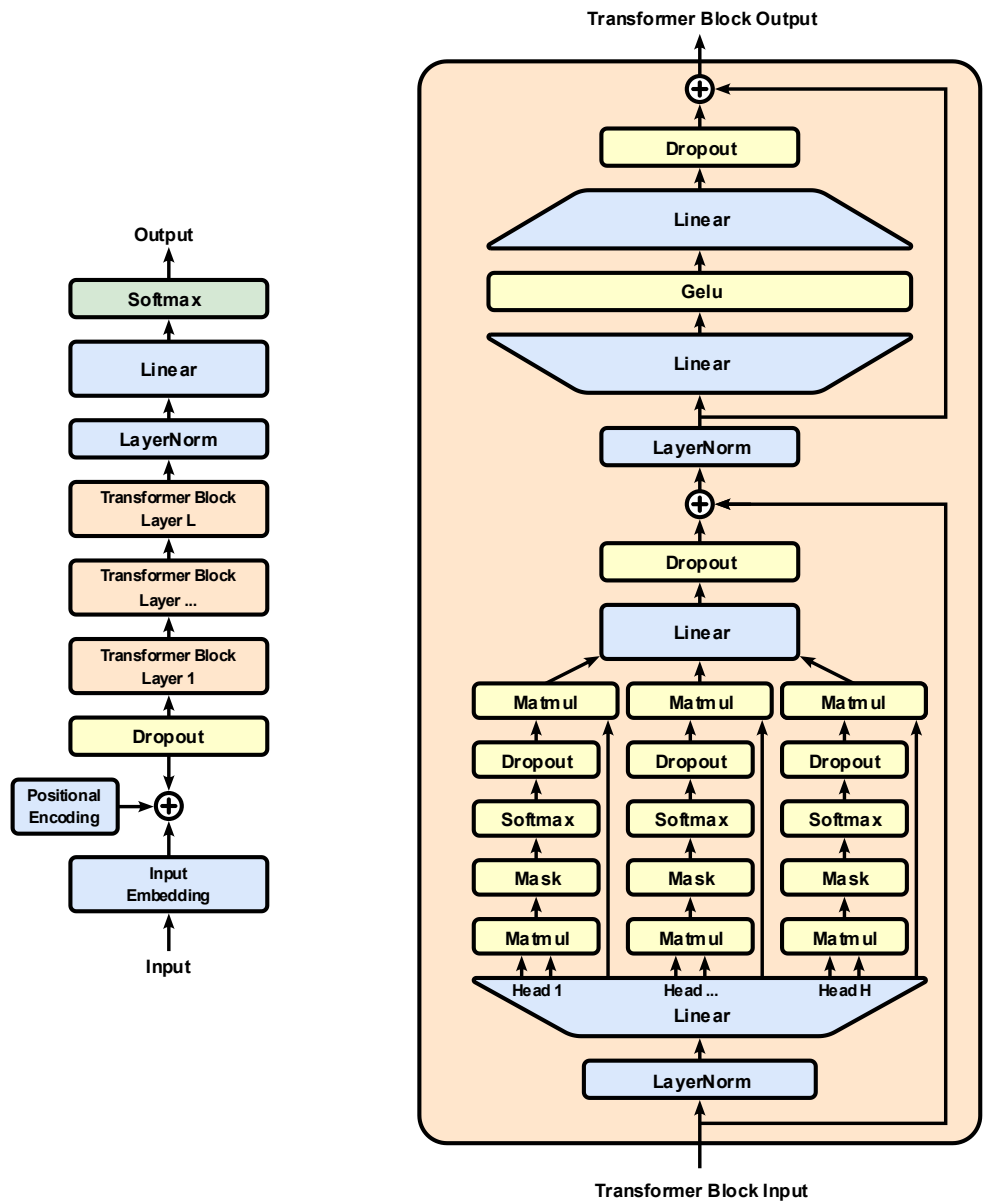
Sparsity & Low-Rankness & Redundancy of PEs

助理教授 纪 焘

taoji@fudan.edu.cn



Attention@GPT



$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V})_m = \frac{\sum_{n=1}^N \text{sim}(\mathbf{q}_m, \mathbf{k}_n) \mathbf{v}_n}{\sum_{n=1}^N \text{sim}(\mathbf{q}_m, \mathbf{k}_n)}.$$

$$\text{sim}(\mathbf{q}_m, \mathbf{k}_n) = \exp(\mathbf{q}_m^\top \mathbf{k}_n / \sqrt{d})$$

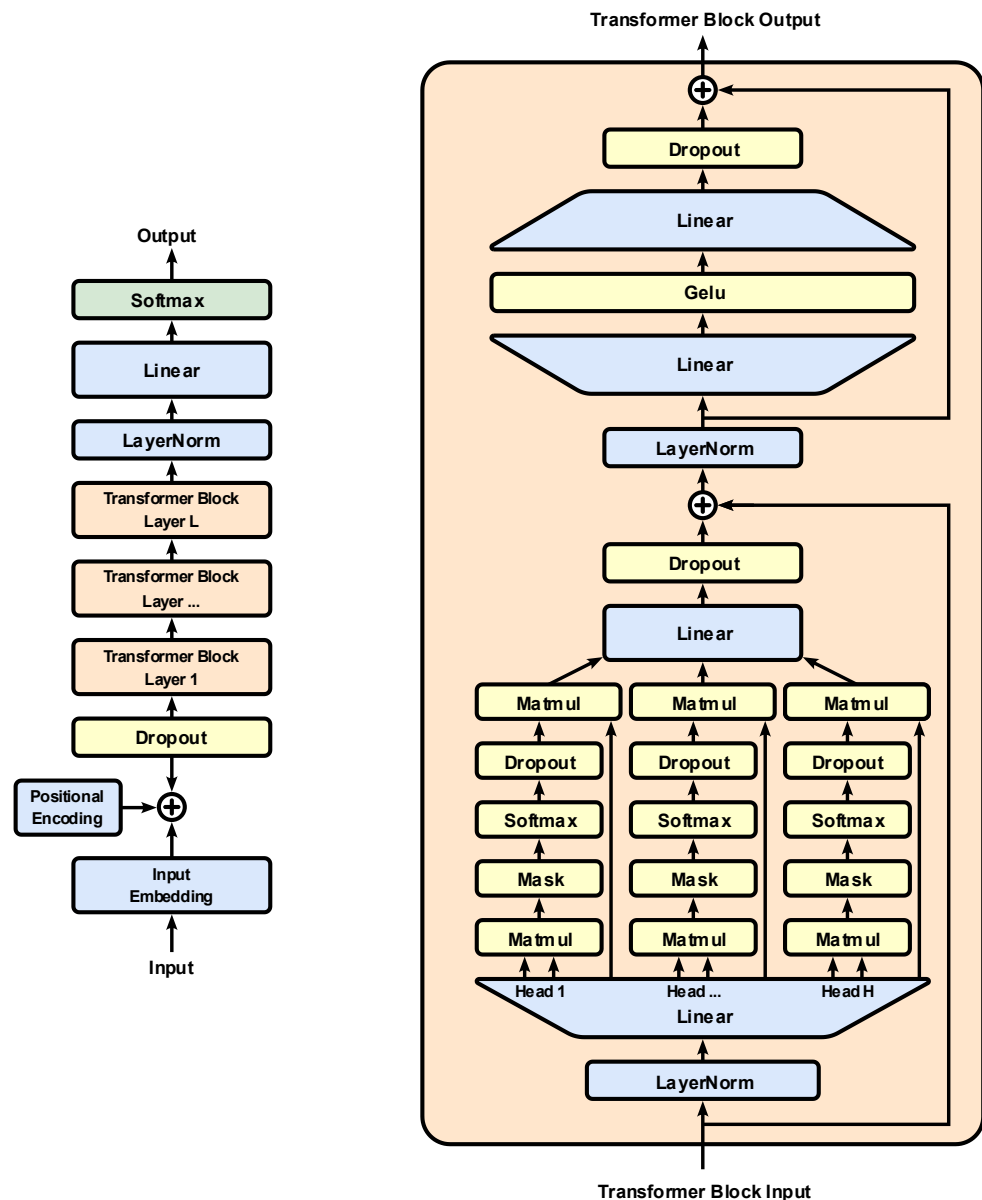
$$\mathbf{q}_m = f_q(\mathbf{x}_q, m)$$

$$\mathbf{k}_n = f_k(\mathbf{x}_k, n)$$

Positional
Encoding

Attention是Transformer的核心组件之

Attention@GPT



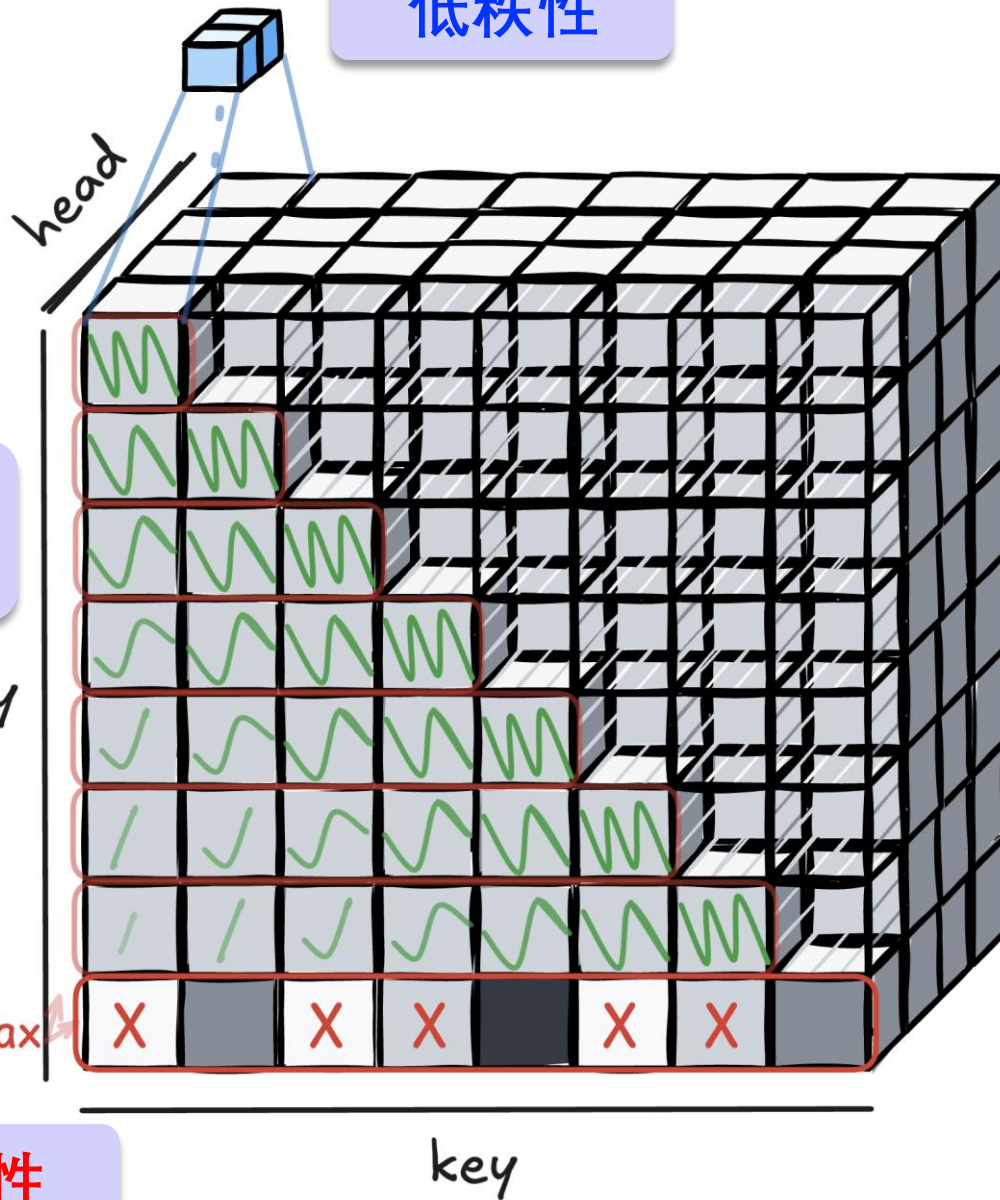
位置冗余

query

softmax

稀疏性

低秩性



Attention计算量分析

评估计算量

毋庸置疑，SA的复杂度是 $\mathcal{O}(n^2)$ ，而FFN的复杂度则是 $\mathcal{O}(n)$ ，如果你直接凭此就想当然地说SA计算量比FFN大，那就错了！

我们知道加法比乘法快很多，所以在估计计算量的时候我们主要计算要做多少次乘法，神经网络里边，主要的运算是矩阵相乘，不难估计按照定义一个 $a \times b$ 的矩阵乘以一个 $b \times c$ 的矩阵要做 abc 次乘法，所以 abc 就是两个矩阵相乘的复杂度了，这是我们估算Transformer复杂度的依据。

设 n 为序列长度， d 为head_size（base版是64）， h 为head的数目（base版是12），那么 hd 就是我们通常说的“hidden_size”（base版是768）。对于SA来说，一开始是 Q, K, V 的投影变换，即 $n \times hd$ 的矩阵乘以 $hd \times hd$ 的矩阵做3次，因此计算量是 $3n(hd)^2$ ；然后是 h 个Attention头的运算，每个头先是 $n \times d$ 的 Q 与 $d \times n$ 的 K^T 相乘得到 $n \times n$ 的Attention矩阵（softmax和归一化的计算量暂且忽略），然后 $n \times n$ 的矩阵与 $n \times d$ 的 V 相乘得到 $n \times d$ 的矩阵，这两步的计算量都是 n^2d ，所以总计算量是 $h(n^2d + n^2d)$ ；最后的输出还有一个投影变换，也是 $n \times hd$ 的矩阵乘以 $hd \times hd$ 的矩阵，计算量是 $n(hd)^2$ 。所以，SA的总计算量是

$$3n(hd)^2 + h(n^2d + n^2d) + n(hd)^2 = 4nh^2d^2 + 2n^2hd \quad (1)$$

至于FFN就比较简单了，它就是两个全连接层，也就是两个矩阵变换（激活函数的计算量也忽略不计），一般的参数设置是：第一层是 $n \times hd$ 的矩阵乘以 $hd \times 4hd$ 的矩阵，第二层就是 $n \times 4hd$ 的矩阵乘以 $4hd \times hd$ 的矩阵。所以总计算量是

$$n \times hd \times 4hd + n \times 4hd \times hd = 8nh^2d^2 \quad (2)$$

这样一来，如果SA的计算量比FFN大，就意味着**单向注意力@MoE⁸/₆₄： $2n > hd$**

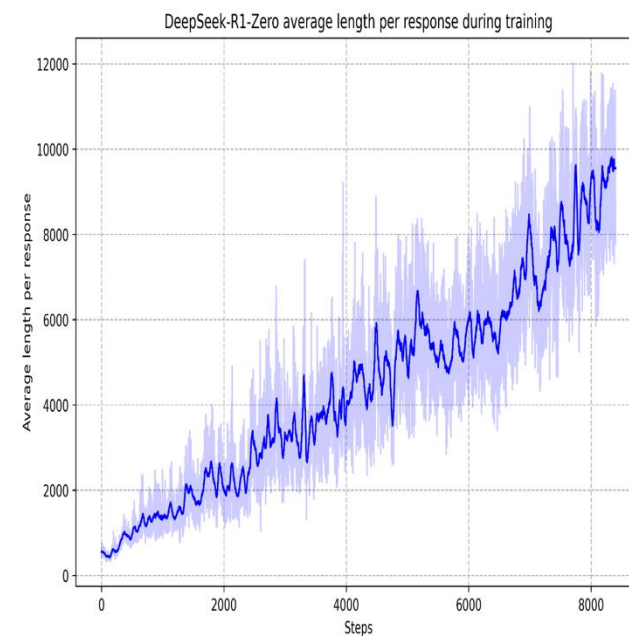
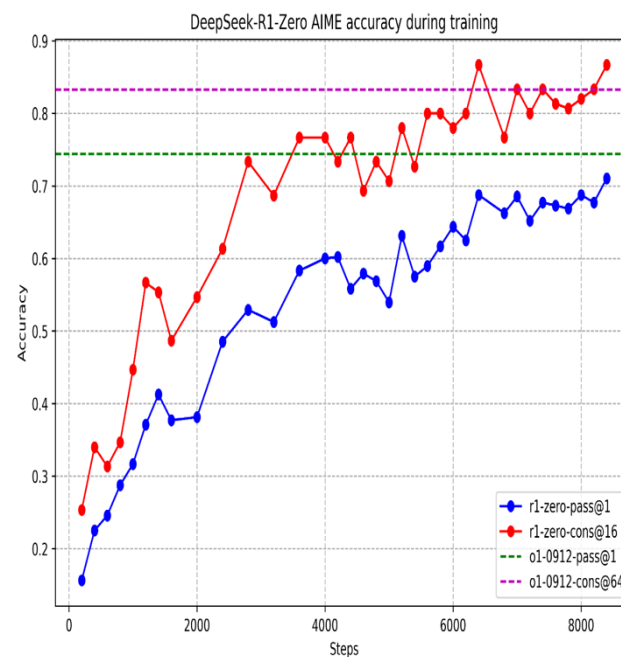
$$4nh^2d^2 + 2n^2hd > 8nh^2d^2 \quad \Leftrightarrow \quad n > 2hd \quad (3)$$

Qwen3 A3B
> 2K

经济高效Attention



**Multi-modal
Scaling
→ Embodied AI...**

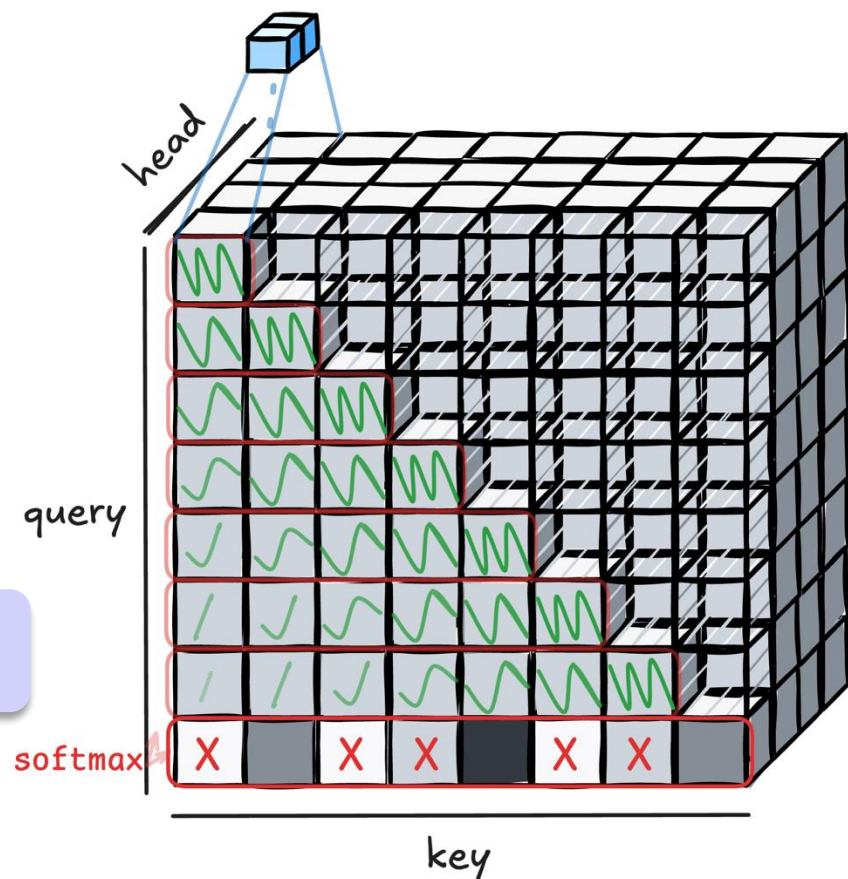


**Test-time Scaling
→ Agent...**

Attn内生的稀疏性

LONGHEADS: Multi-Head Attention is Secretly a Long Context Processor

Yi Lu^{1*}, Xin Zhou^{1*}, Wei He¹, Jun Zhao¹,
Tao Ji^{1†}, Tao Gui^{2†}, Qi Zhang^{1†}, Xuanjing Huang^{1,3}



Softmax $\approx$ Argmax

稀疏性

Full vs. Restricted vs. LongHeads

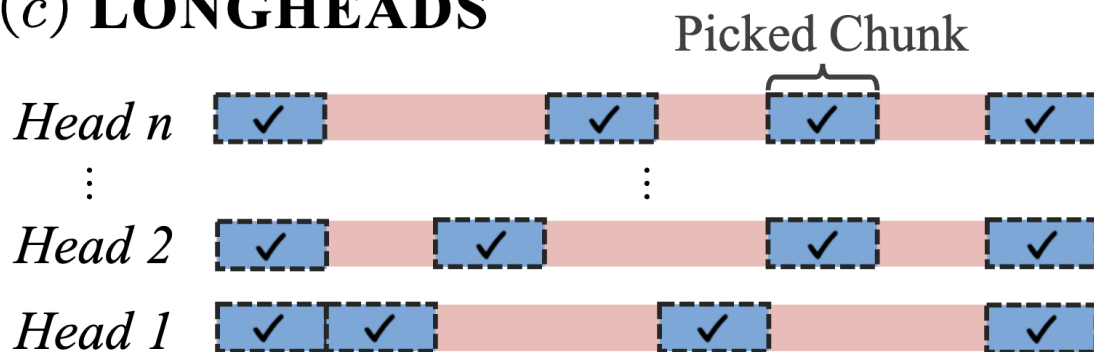
(a) **Full Attention**



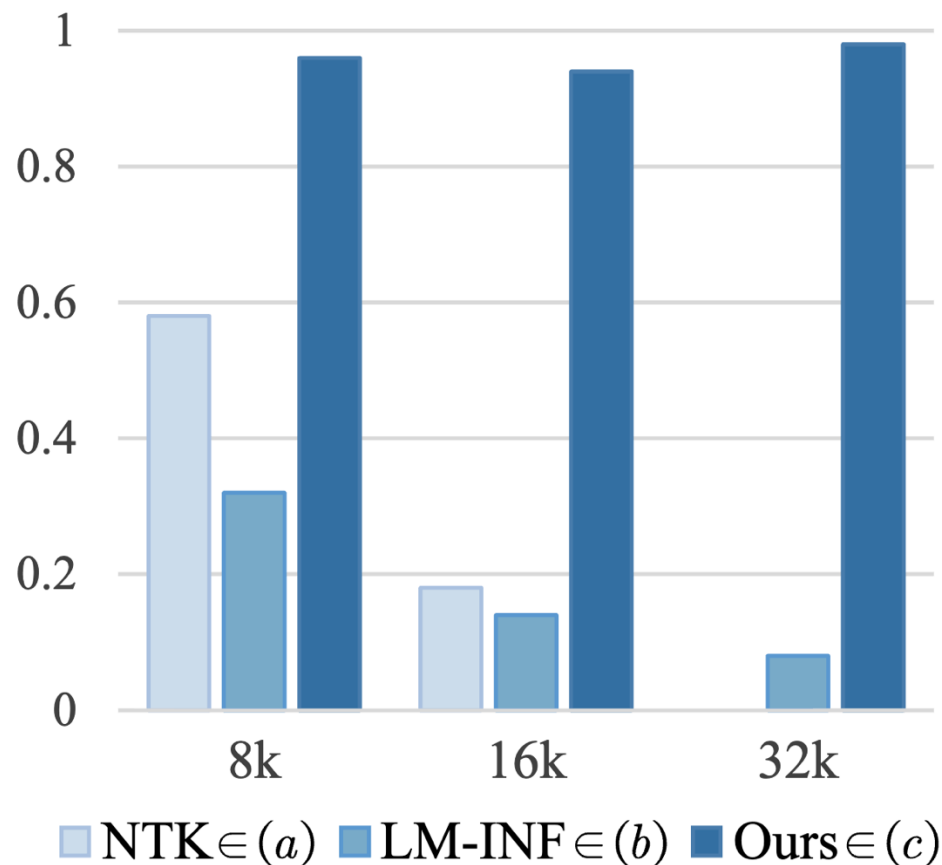
(b) **Restricted Attention**



(c) **LONGHEADS**



Accuracy on Passkey Retrieval



Longheads: 上下文分块+稀疏选择推理

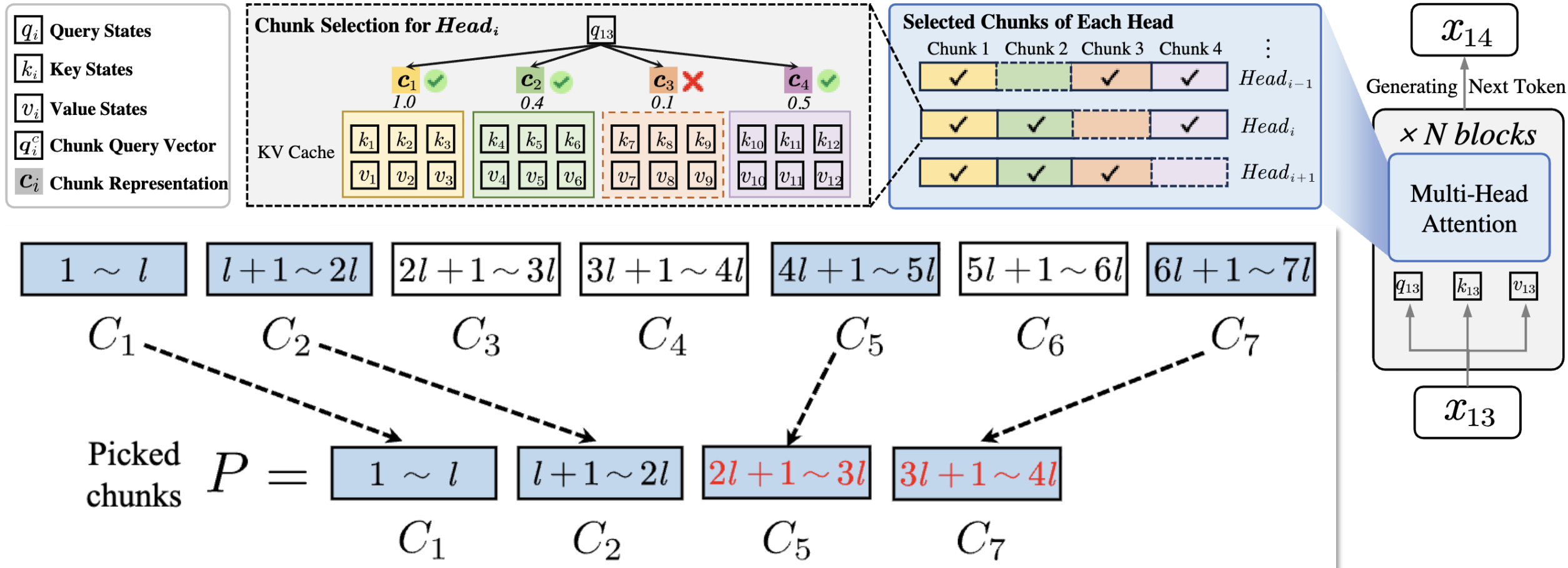
$O(M^2)$

$O(N^2) \rightarrow O((k * C)^2) \rightarrow$

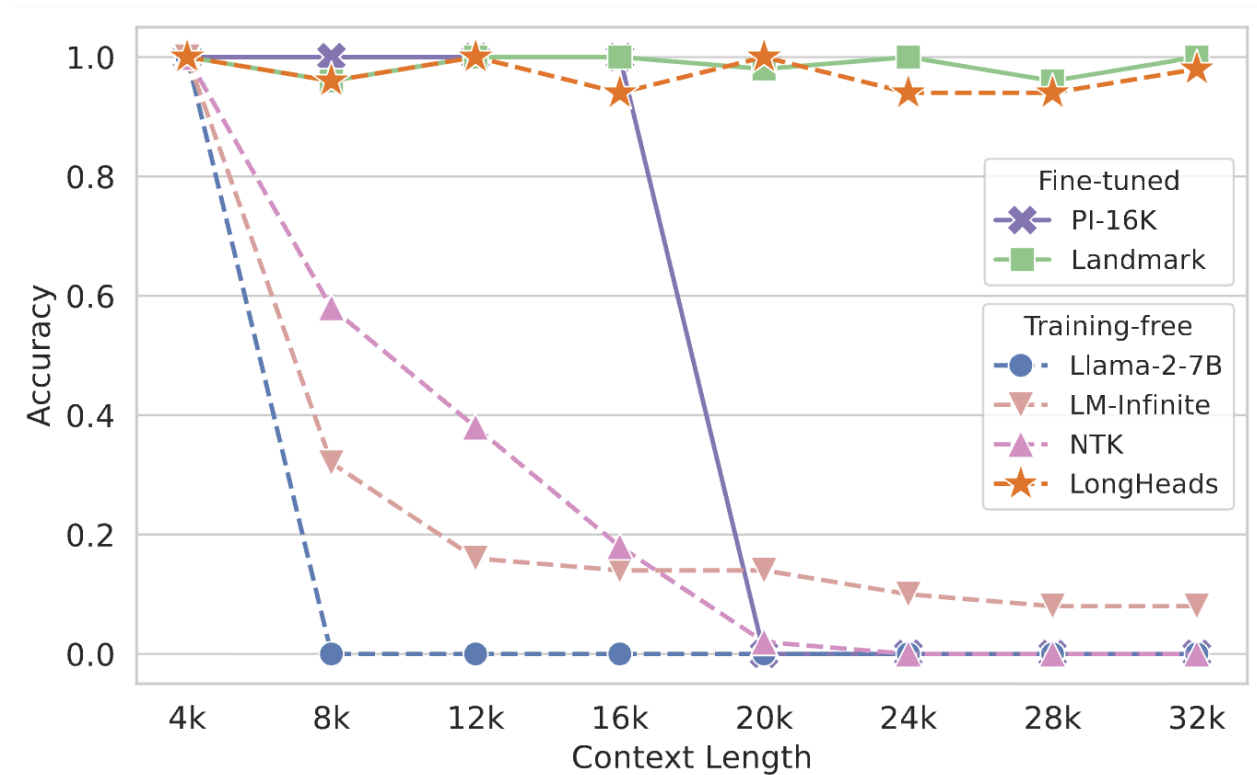
LongHeads: 上下文分块+稀疏选择

Training-free选择策略

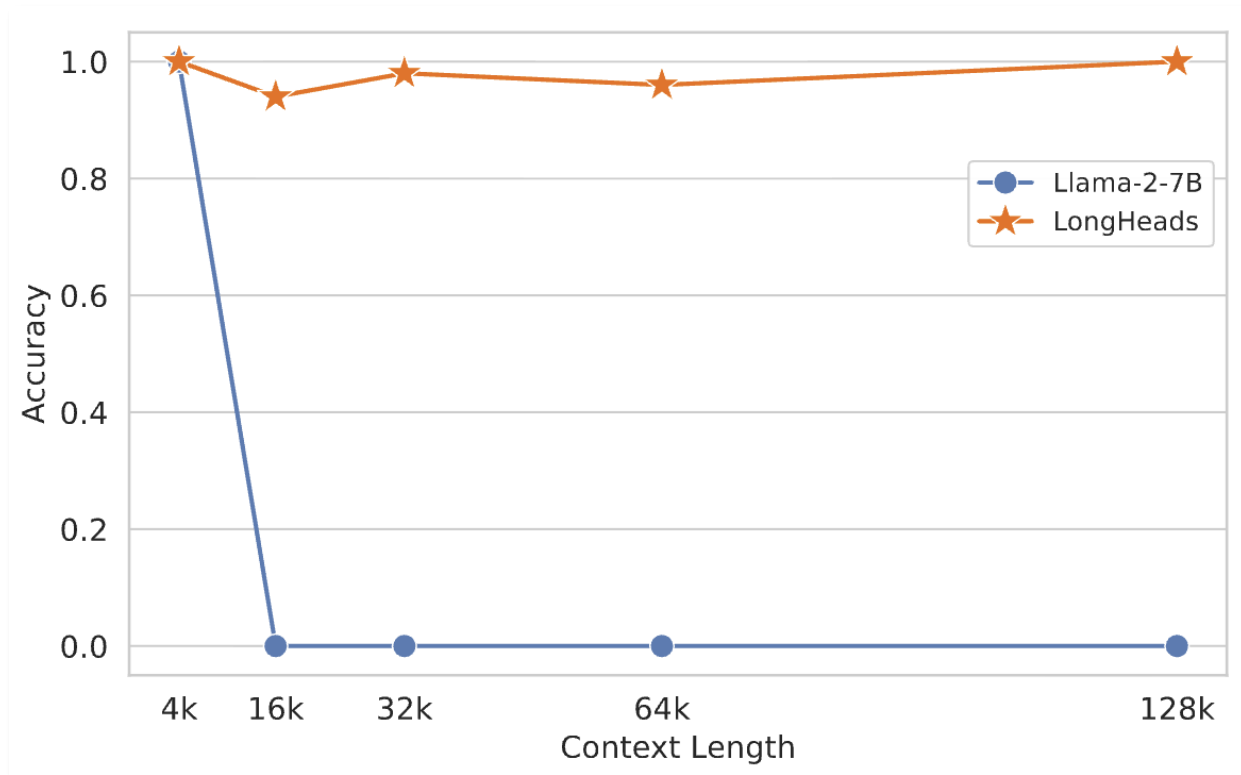
未选择的上下文块 → score=0



LongHeads: 上下文分块+稀疏选择



Training-free与微调方法相当



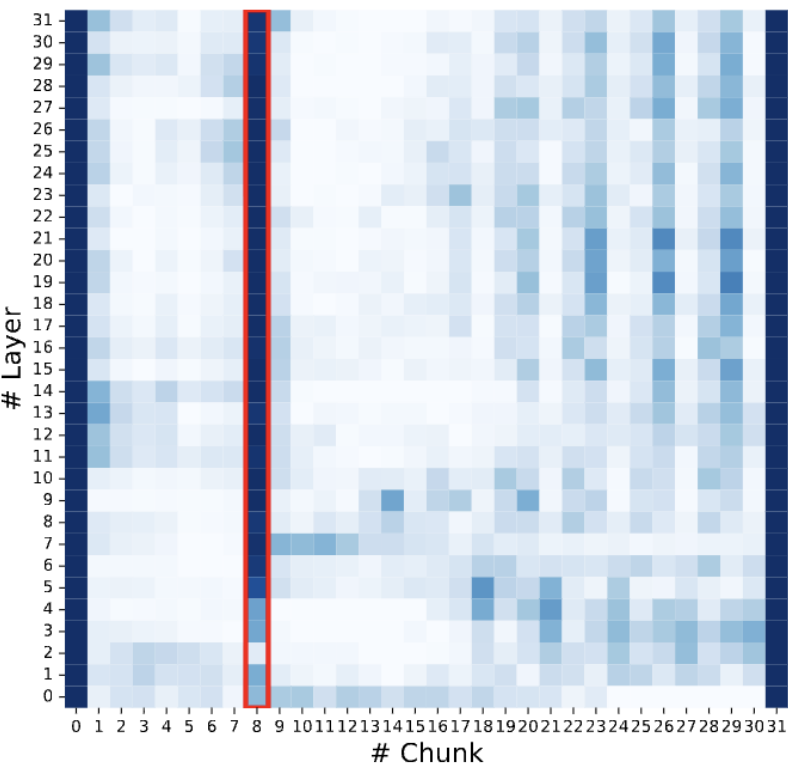
4K基座外推到128K

LongHeads: 上下文分块+稀疏选择

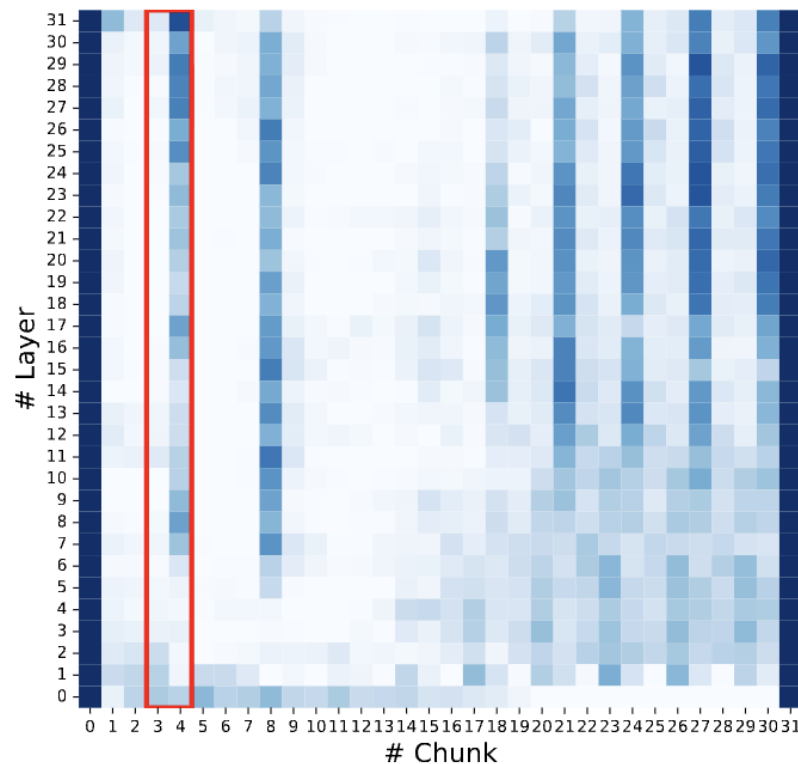
Method	FT Tokens	Single-Doc QA			Multi-Doc QA			Summarization			Few-shot Learning			Synthetic		Code		Avg.
		NQA	Qspr.	MulFi	HQA	WMQA	Musq.	GRpt	QMSM	MulN	TREC	TriQA	SMSM	PsgC	PsgR	Lcc	Repo	
Full Attention																		
NTK	-	16.47	29.62	31.42	31.31	28.75	10.20	22.70	17.65	6.31	64.67	77.36	37.95	3.99	5.12	65.64	52.97	31.38
PI-16k	0.85B	21.37	31.78	36.67	37.56	27.47	15.98	13.55	20.69	1.18	63.00	89.24	25.64	5.67	11.33	67.05	56.02	32.76
Restricted Attention																		
LM-Infinite	-	14.34	20.75	26.18	20.37	20.08	5.87	16.70	7.01	2.28	54.67	76.69	15.64	4.30	7.00	62.90	52.74	25.47
Landmark	0.80B	11.35	23.91	20.96	26.95	26.25	5.22	17.74	19.15	9.84	42.67	80.73	35.45	5.73	7.00	59.74	42.76	27.22
LONGHEADS	-	14.51	21.58	30.32	30.07	25.28	9.15	24.74	20.26	6.30	55.00	83.26	34.27	2.45	9.39	65.01	50.65	30.14
w/ NTK init	-	16.48	28.63	31.36	31.19	28.67	13.54	22.85	17.63	6.38	65.33	77.49	38.07	4.32	4.97	65.56	52.87	31.58
w/ PI init	0.85B	21.43	31.78	36.64	37.63	27.33	15.98	13.36	20.57	1.30	63.00	89.57	25.86	5.67	11.33	66.93	48.96	32.33
Extend to 32k																		
NTK	-	5.74	29.05	31.39	28.98	27.03	9.34	22.00	15.13	5.40	64.67	48.34	34.50	3.89	4.85	57.54	45.29	27.07
PI-16k	0.85B	8.43	30.15	35.20	29.47	24.72	1.74	13.23	12.59	1.30	55.00	66.15	19.16	5.42	11.33	33.21	27.21	23.39
LM-Infinite	-	10.87	20.58	26.19	19.48	20.40	16.52	5.26	2.51	6.14	55.00	82.78	11.26	4.30	6.67	64.88	56.02	25.55
Landmark	0.80B	13.88	23.69	21.06	28.04	25.78	11.52	17.70	19.11	10.68	41.00	77.15	35.61	5.70	7.00	58.22	40.97	27.32
LONGHEADS	-	13.38	21.81	30.33	29.59	24.90	11.48	27.43	19.87	6.07	55.00	81.15	33.56	2.79	10.06	63.75	47.97	29.95

效率↔Restricted, 性能↔Full

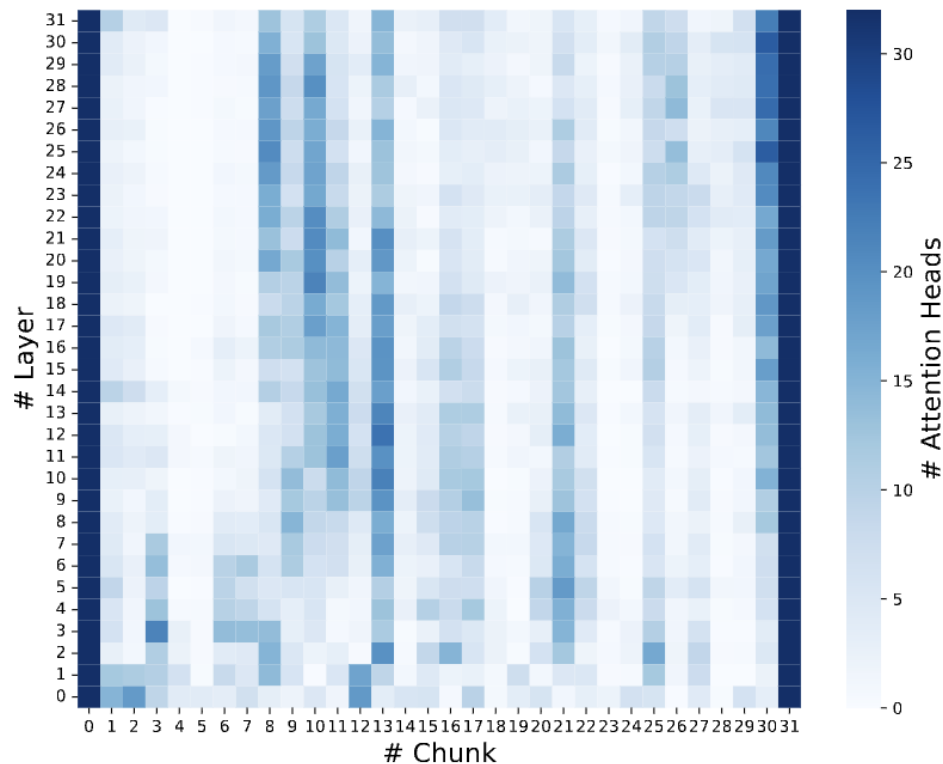
LongHeads: 上下文分块+稀疏选择



(a) Passkey Retrieval Task (Success)



(b) Passkey Retrieval Task (Fail)



(c) Summarization Task

效率 ↔ Restricted, 性能 ↔ Full

LongHeads: 上下文分块+稀疏选择

Method	4k			16k			32k			128k		
	Time(s)	Mem(GB)	Acc(%)	Time(s)	Mem(GB)	Acc(%)	Time(s)	Mem(GB)	Acc(%)	Time(s)	Mem(GB)	Acc(%)
Llama2	0.03	18.8	100	-	OOM	-	-	OOM	-	-	OOM	-
Flash Atten	0.03	17.2	100	0.11	30.8	0	0.21	49.0	0	-	OOM	-
PI	0.03	17.2	100	0.11	30.8	100	0.21	49.0	0	-	OOM	-
NTK	0.04	17.2	100	0.11	30.8	12	0.21	49.0	0	-	OOM	-
YaRN	0.03	17.2	100	0.11	30.8	92	0.21	49.0	0	-	OOM	-
Landmark	0.17	18.8	100	0.45	31.2	96	0.87	48.0	100	-	OOM	-
ChunkLLaMa	0.07	17.4	94	0.12	31.2	80	0.16	49.8	64	-	OOM	-
LM-Infinite	0.05	17.2	32	0.10	38.6	14	0.17	65.6	8	-	OOM	-
LongHeads	0.03	19.0	96	0.08	29.9	94	0.11	47.1	98	4.14*	42.3*	100*

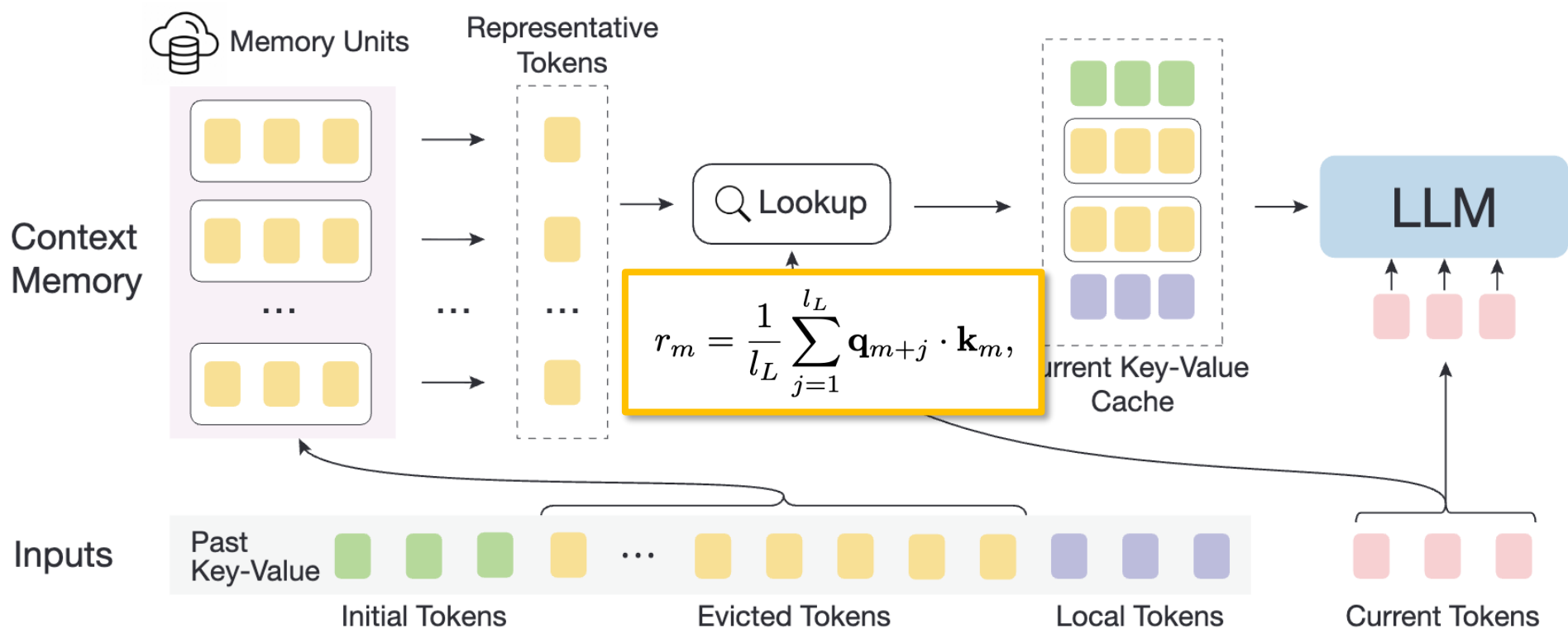
LongHeads显存仅随Chunk数增长，结合cpu offloading单卡可推128K

LongHeads: 上下文分块+稀疏选择

Method Setting	LongBench Avg.
LONGHEADS	30.14
- Random Selection	28.77
- Last K Selection	26.22
- w/o First Selection	14.06
- Fix Head	29.46
- Fix Layer	28.78
- Fix Head & Layer	28.72
- Number of Chunks $K = 8$	29.09
- Number of Chunks $K = 4$	26.64
- Chunk Size $l = 512$	29.95
- Chunk Size $l = 128$	29.35

InfLLM: Training-Free Long-Context Extrapolation for LLMs with an Efficient Context Memory

Chaojun Xiao^{1*}, Pengle Zhang^{1*}, Xu Han^{1†}, Guangxuan Xiao²,
Yankai Lin³, Zhengyan Zhang¹, Zhiyuan Liu^{1†}, Maosong Sun¹



LongHeads inspired MoBA & DeepSeek-NSA

 MoBA: Mixture of Block Attention

TECHNICAL REPORT

closely related to MoBA is Longheads ([Y. Lu et al. 2024](#)) which can be viewed as MoBA with a top-1 gating network, meaning that each query selects the most relevant KV blocks for attention.



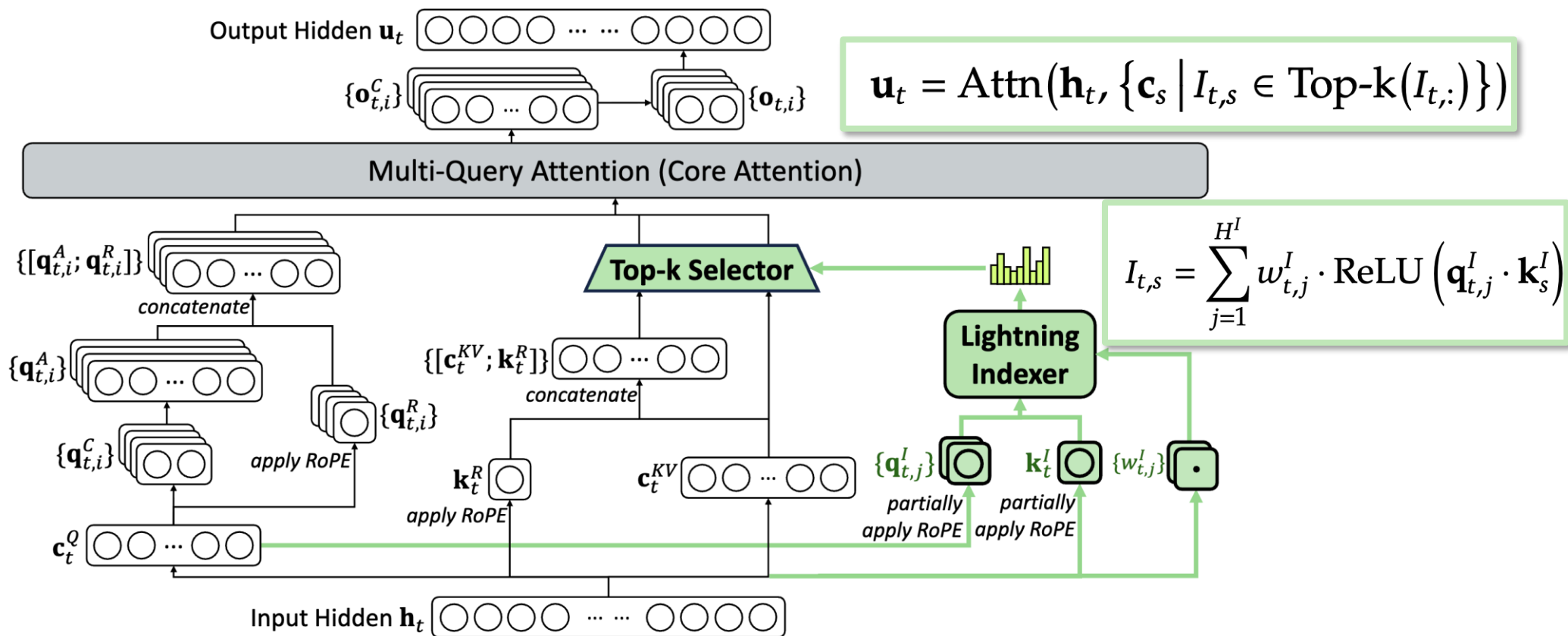
Native Sparse Attention: Hardware-Aligned and Natively Trainable Sparse Attention

Jingyang Yuan^{*1,2}, Huazuo Gao¹, Damai Dai¹, Junyu Luo², Liang Zhao¹, Zhengyan Zhang¹, Zhenda Xie¹,
Y. X. Wei¹, Lean Wang¹, Zhiping Xiao³, Yuqing Wang¹, Chong Ruan¹, Ming Zhang², Wenfeng Liang¹,
Wangding Zeng¹

DeepSeek-DSA

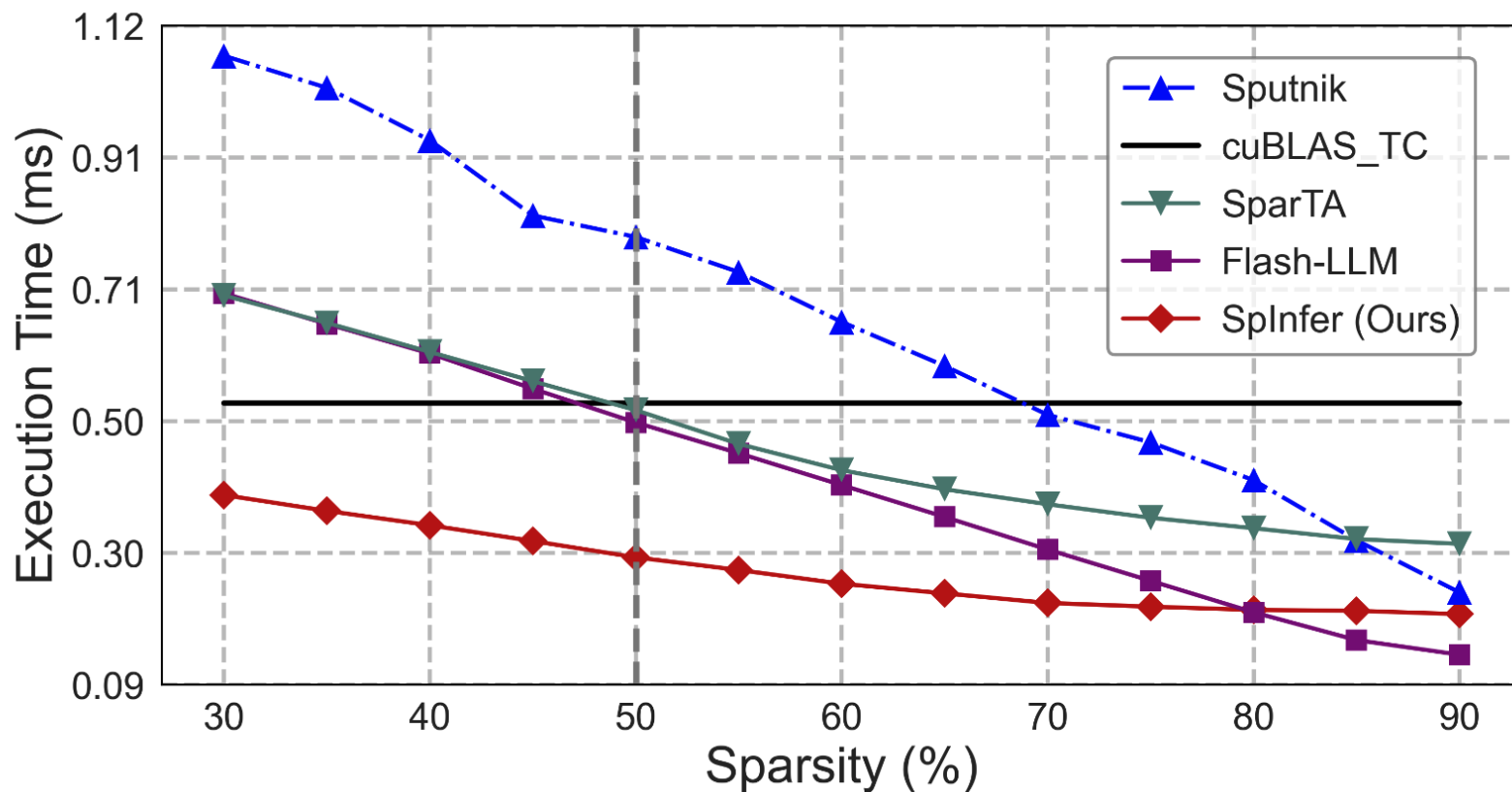
DeepSeek-V3.2-Exp: Boosting Long-Context Efficiency with DeepSeek Sparse Attention

DeepSeek-AI



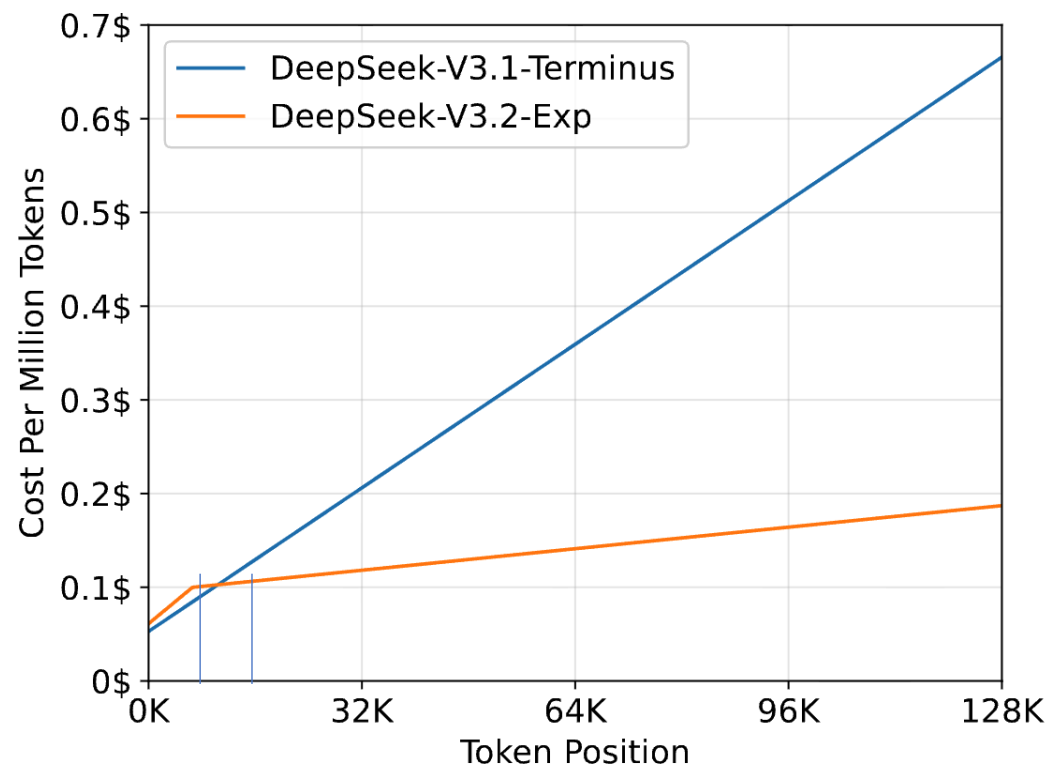
DeepSeek-DSA：非结构化稀疏

EuroSys '25, March 30-April 3, 2025, Rotterdam, Netherlands

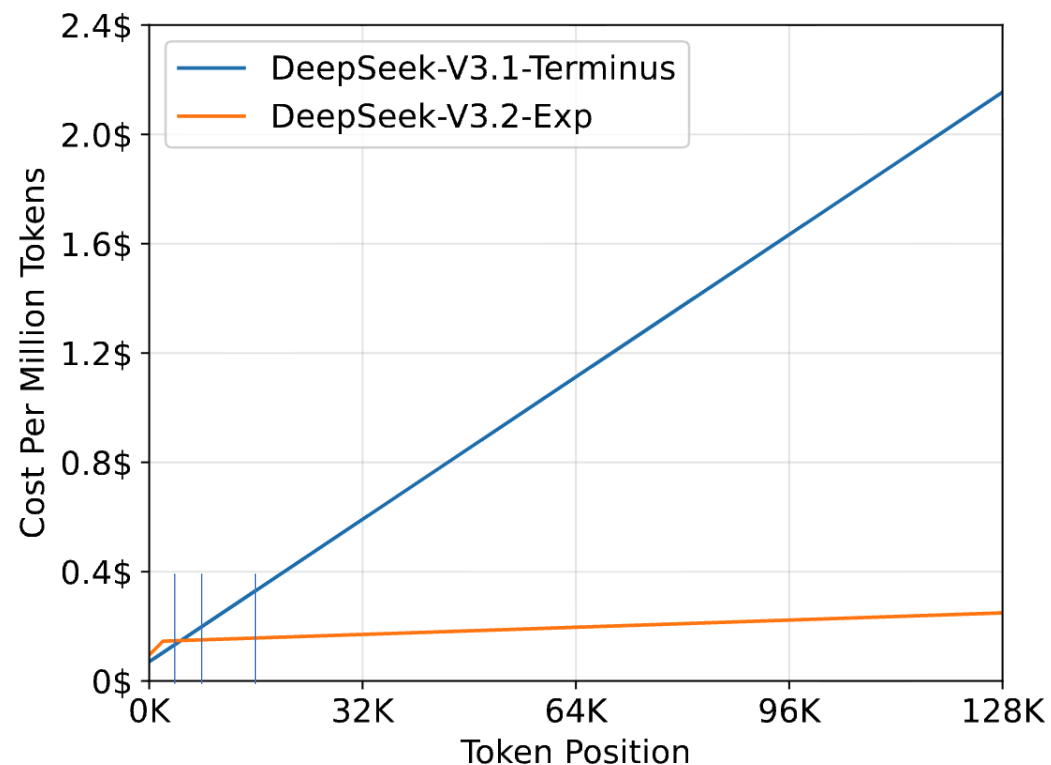


非结构化稀疏收益刚刚体现 (@A卡)

DeepSeek-DSA: 非结构化稀疏



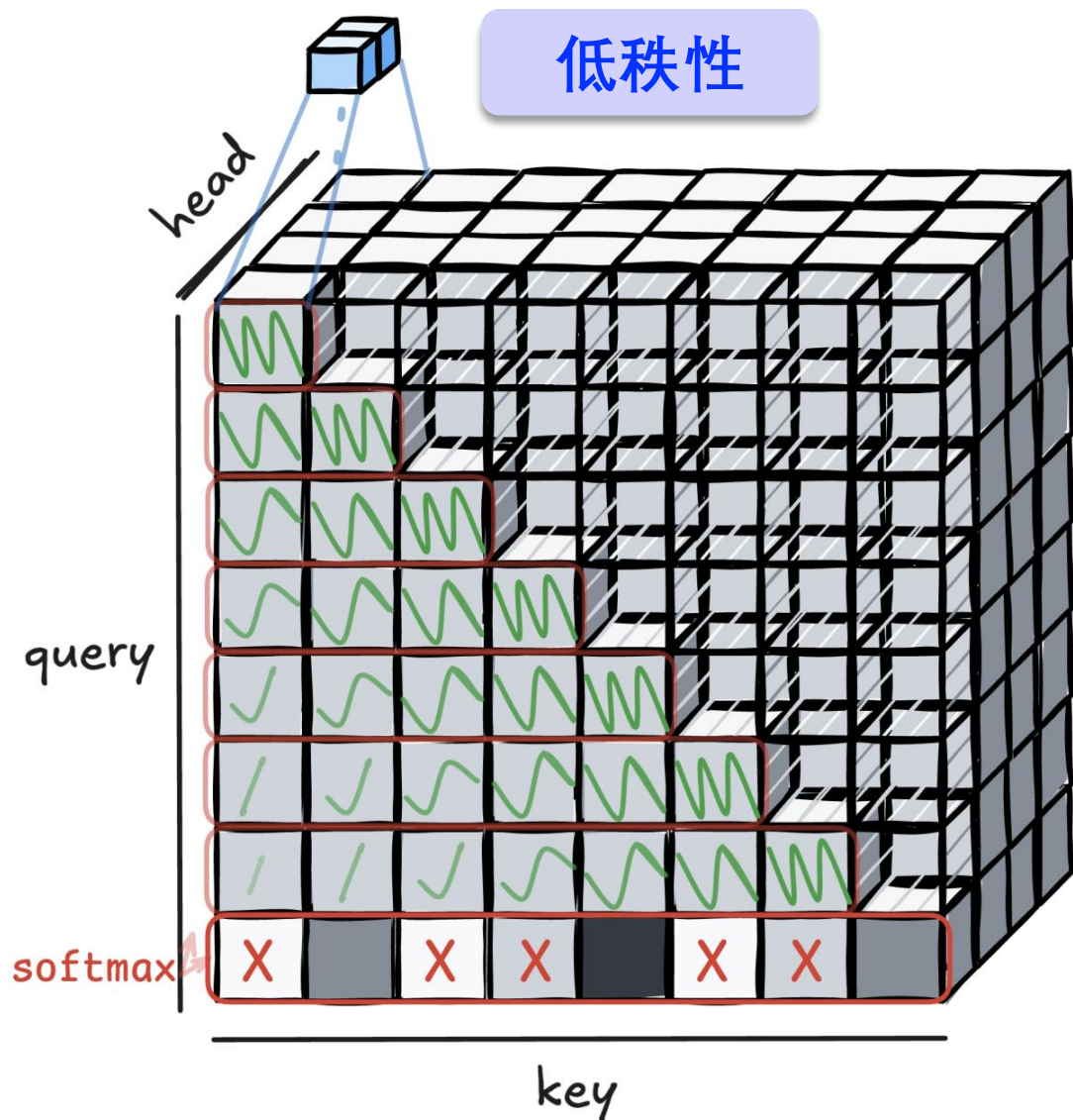
(a) Prefilling



(b) Decoding

2K@4K $\approx$ 50% 2K@32K $\approx$ 94%

Attn内生的低秩性



Linear Transformation可被分解

Attn内生的低秩性

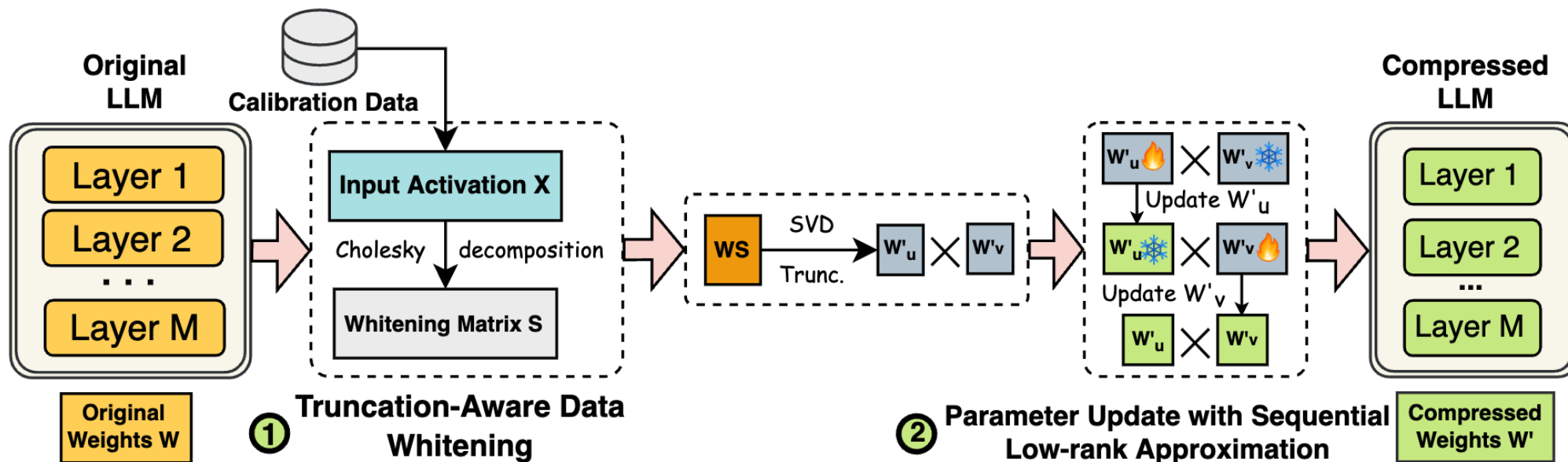


SVD-LLM: TRUNCATION-AWARE SINGULAR VALUE DECOMPOSITION FOR LARGE LANGUAGE MODEL COMPRESSION

Xin Wang¹ Yu Zheng² Zhongwei Wan¹ Mi Zhang¹

¹The Ohio State University ²Michigan State University

<https://github.com/AIoT-MLSys-Lab/SVD-LLM>



Attn内生的低秩性

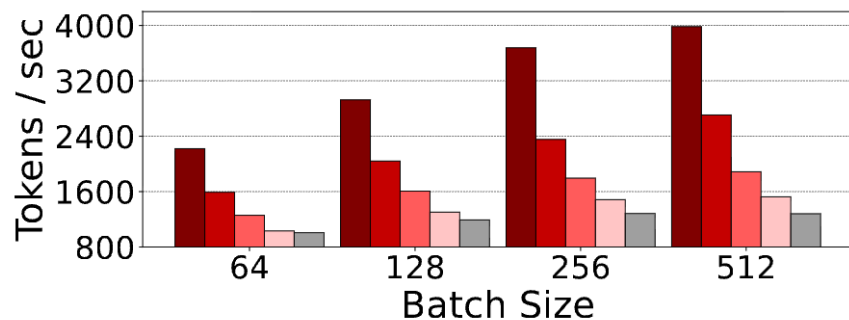
RATIO (MEM.)	METHOD	WikiText-2↓	C4↓	Openb.	ARC_e	WinoG.	HellaS.	PIQA	MathQA	Average↑	TruthfulQA↑	GSM8K↑
0% (13.5 GB)	Original	5.68	7.34	0.34	0.75	0.70	0.57	0.79	0.27	0.57	0.30	0.09
20% (10.2 GB)	SVD	20061	18800	0.05	0.04	0.01	0.03	0.02	0.03	0.03	0.00	0.00
	FWSVD	1727	1511	0.09	0.11	0.05	0.08	0.10	0.05	0.08	0.00	0.00
	ASVD	11.14	15.93	0.29	0.53	0.64	0.41	0.68	0.17	0.45	0.21	0.04
40% (7.76 GB)	SVD-LLM (W)	7.94 (↓29%)	15.84 (↓1%)	0.31	0.62	0.61	0.45	0.71	0.21	0.49 (↑9%)	0.26 (+0.05)	0.05 (+0.01)
	SVD-LLM	7.73 (↓31%)	12.23 (↓23%)	0.33	0.67	0.69	0.55	0.79	0.26	0.55 (↑22%)	0.28 (+0.07)	0.08 (+0.04)
	ASVD	1407	1109	0.08	0.11	0.09	0.08	0.13	0.08	0.10	0.01	0.00
60% (5.35 GB)	SVD	105474	106976	0.01	0.03	0.01	0.00	0.01	0.02	0.01	0.00	0.00
	FWSVD	32194	29292	0.06	0.02	0.01	0.01	0.02	0.03	0.03	0.00	0.00
	ASVD	57057	43036	0.05	0.04	0.06	0.09	0.08	0.05	0.06	0.00	0.00
80% (2.58 GB)	SVD-LLM (W)	66.62 (↓99%)	471.83 (↓99%)	0.10	0.05	0.17	0.10	0.21	0.04	0.11 (↑83%)	0.01 (+0.01)	0.00 (+0.00)
	SVD-LLM	15.00 (↓99%)	26.26 (↓99%)	0.18	0.42	0.44	0.31	0.35	0.12	0.30 (↑400%)	0.14 (+0.14)	0.04 (+0.04)
	ASVD	80425	67927	0.04	0.03	0.03	0.02	0.01	0.01	0.03	0.00	0.00
20% (10.2 GB)	SVD	687291	708243	0.00	0.01	0.02	0.01	0.01	0.00	0.01	0.00	0.00
	FWSVD	96872	89243	0.01	0.02	0.00	0.01	0.01	0.00	0.01	0.00	0.00
	ASVD	80425	67927	0.04	0.03	0.03	0.02	0.01	0.01	0.03	0.00	0.00
40% (7.76 GB)	SVD-LLM (W)	1349 (↓98%)	6224 (↓91%)	0.07	0.03	0.04	0.02	0.07	0.01	0.04 (↑33%)	0.00 (+0.00)	0.00 (+0.00)
	SVD-LLM	31.79 (↓99%)	43.71 (↓99%)	0.11	0.23	0.21	0.14	0.17	0.08	0.16 (↑433%)	0.04 (+0.04)	0.02 (+0.02)
	ASVD	80425	67927	0.04	0.03	0.03	0.02	0.01	0.01	0.03	0.00	0.00

压缩率可控

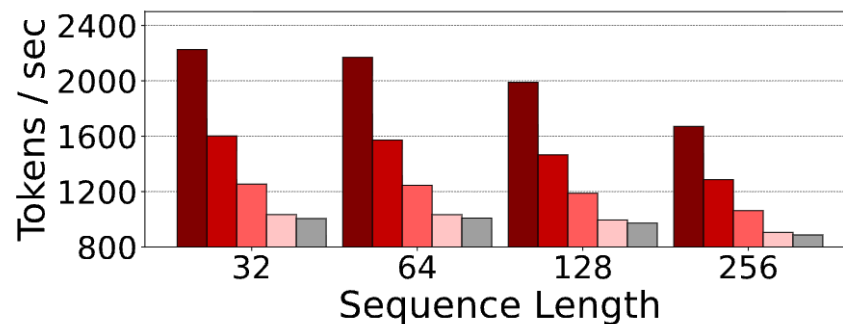
	OPT-6.7B		LLAMA 2-7B		MISTRAL-7B		VICUNA-7B	
METHOD	Perplexity↓	Accuracy↑	Perplexity↓	Accuracy↑	Perplexity↓	Accuracy↑	Perplexity↓	Accuracy↑
Original	10.86	0.52	5.47	0.57	5.25	0.61	6.78	0.56
SVD	66275	0.03	18192	0.09	159627	0.03	18644	0.05
FWSVD	14559	0.06	2360	0.12	6357	0.08	2758	0.09
ASVD	82.00	0.32	10.10	0.36	13.72	0.32	16.23	0.33
SVD-LLM (W)	16.04 (↓80%)	0.41 (↑28%)	8.50 (↓16%)	0.53 (↑47%)	10.21 (↓26%)	0.42 (↑24%)	8.41 (↓48%)	0.51 (↑55%)
SVD-LLM	14.47 (↓82%)	0.49 (↑53%)	7.73 (↓23%)	0.54 (↑50%)	7.47 (↓45%)	0.55 (↑72%)	7.43 (↓54%)	0.54 (↑64%)

20%压缩比
不影响性能

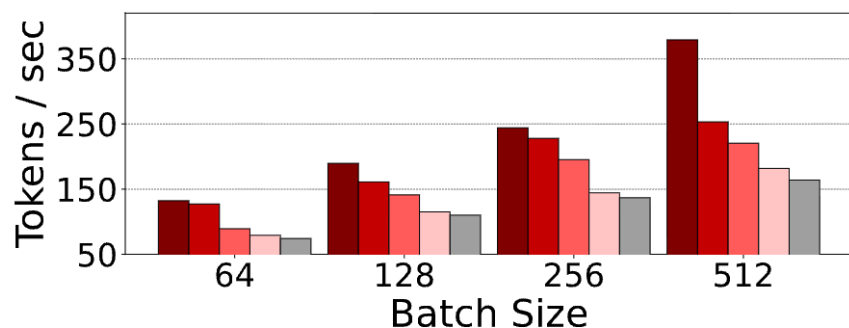
Attn内生的低秩性



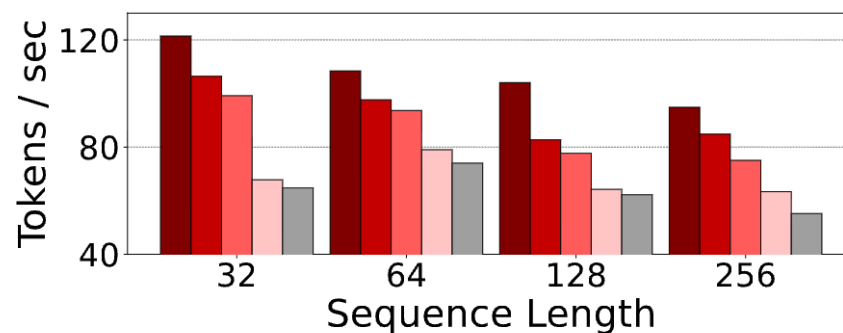
(a) Varying Batch Size on GPU



(b) Varying Sequence Length on GPU



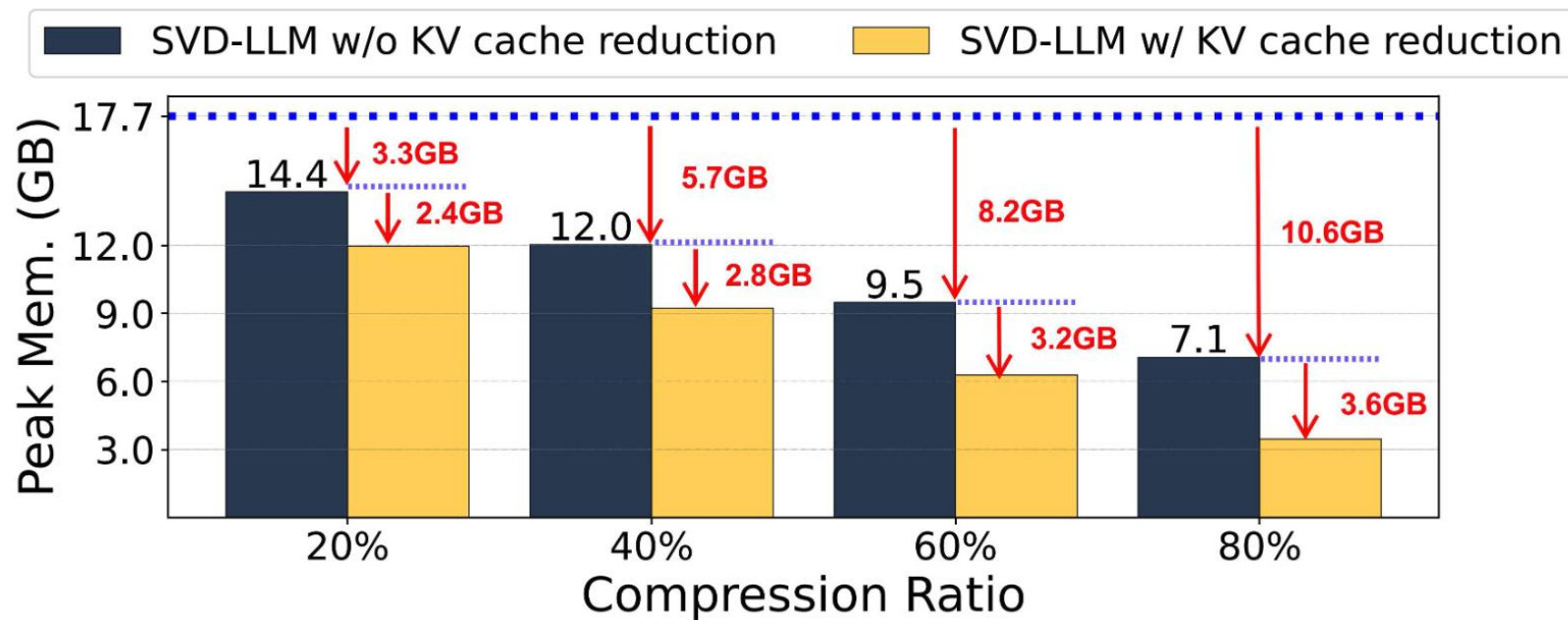
(c) Varying Batch Size on CPU



(d) Varying Sequence Length on CPU

压缩率越高，推理速度越快

Attn内生的低秩性



显存节省显著

Attn内生的低秩性

SVD-LLM V2: Optimizing Singular Value Truncation for Large Language Model Compression

Xin Wang Samiul Alam Zhongwei Wan Hui Shen Mi Zhang
The Ohio State University
{wang.15980, alam.140, wan.512, shen.1780, mizhang.1}@osu.edu
<https://github.com/AIoT-MLSys-Lab/SVD-LLM>

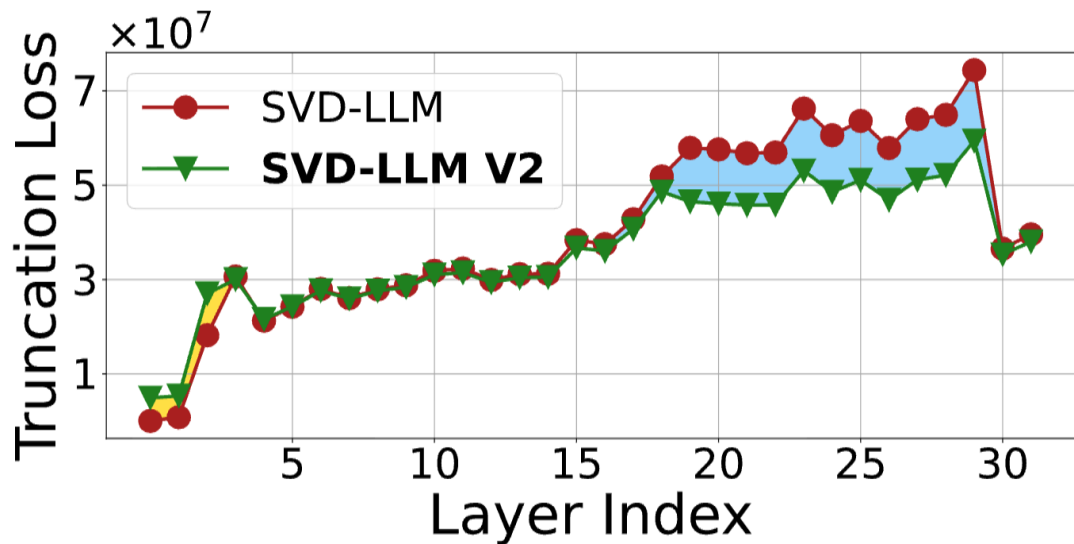
	Compression Ratio	Truncation Loss	PPL
SVD-LLM	Homogeneous	L = 0.8961	P = 11.8 ❌
SVD-LLM V2	Heterogeneous	L = 0.7351	P = 8.01 ✅

动态压缩率

改进近似算法

层动态压缩率

高误差→压缩率减少



低误差→压缩率增大

Algorithm 1 Pseudocode of Heterogeneous Compression Ratio Allocation in SVD-LLM V2

Input: M : Original LLM
 x : Input activation
 R : Target compression ratio

Output: R_d : A list of allocated compression ratios

```
1: procedure RATIO_ALLOCATION( $M, S, R$ )
2:    $G \leftarrow \mathbf{Group}(M)$   $\triangleright$  Group the weight by types
3:    $R_d \leftarrow \emptyset$   $\triangleright$  Initialize the compression ratio list
4:   for  $g$  in  $G$  do
5:      $L_G \leftarrow \emptyset$   $\triangleright$  Initialize the loss list in the group
6:     for  $w$  in  $g$  do
7:        $L_{min} \leftarrow \mathbf{Theoretical\_Loss}(w, x, R)$ 
8:        $L_G \leftarrow L_G \cup L_{min}$ 
9:     end for
10:     $L_G \leftarrow 1/\mathbf{Log}(L_G)$   $\triangleright$  Normalize  $L_G$ 
11:     $r \leftarrow \mathbf{Len}(L_G) \times R \times L_{min}/\mathbf{Sum}(L_G)$ 
12:     $R_d \leftarrow R_d \cup r$   $\triangleright$  Append  $r$  to the list  $R_d$ 
13:  end for
14:  return  $R_d$ 
15: end procedure
```

激活感知的低秩近似

$$\begin{aligned} L^2 &= \|WX - W'X\|_F^2 \\ &= \|WSS^{-1}X - U_{ws} \times \text{Trunc.}(S_{ws}) \times V_{ws} \times S^{-1}X\|_F^2 \\ &= \|(WS - U_{ws} \times \text{Trunc.}(S_{ws}) \times V_{ws})S^{-1}X\|_F^2 \\ &= \|WS - U_{ws} \times \text{Trunc.}(S_{ws}) \times V_{ws}\|_F^2 \\ &= \|\text{SVD}(WS)\|_F^2 = \|\text{SVD}(W \times U_x \times S_x)\|_F^2 \\ &= \|\text{SVD}(W \times U_x \times S_x \times V_x)\|_F^2 \\ &= \|\text{SVD}(WX)\|_F^2 = L_{min}^2 \end{aligned}$$

Algorithm 2 Pseudocode of Loss-optimized Weight Truncation in SVD-LLM V2

Input: W : Original weight matrix

X : Input activation

R : Target compression ratio

Output: W' : Compressed weight matrix

- 1: **procedure** WEIGHT_TRUNCATION(W, X, R)
 - 2: $S \leftarrow XX^T$ ▷ Construct matrix S from X
 - 3: $U_s, S_s, V_s \leftarrow \text{SVD}(S)$ ▷ Perform SVD on matrix S
 - 4: $D \leftarrow W \times U_s \times \sqrt{S_s}$ ▷ Construct matrix D
 - 5: $U_{ws}, S_{ws}, V_{ws} \leftarrow \text{SVD}(D)$ ▷ Perform SVD on matrix D
 - 6: $T_s \leftarrow \text{Truncate}(S_{ws}, R)$ ▷ Perform SVD truncation on matrix S_{ws} based on compression ratio R
 - 7: $W' \leftarrow U_{ws} \times T_s \times S_s^{-1} \times U_s^{-1}$ ▷ Construct W'
 - 8: **return** W'
 - 9: **end procedure**
-

SVDLLM V2 vs. V1

	METHOD	WikiText-2↓	C4↓	Openb.	ARC_e	WinoG.	HellaS.	PIQA	MathQA	Average↑	TruthfulQA↑	GSM8K↑
LLaMA-7B	Original	5.68	7.34	0.34	0.75	0.70	0.57	0.79	0.27	0.57	0.30	0.09
	FWSVD	1727	1511	0.09	0.11	0.05	0.08	0.10	0.05	0.08	0.00	0.00
	ASVD	11.14	15.93	0.29	0.53	0.64	0.41	0.68	0.17	0.45	0.21	0.04
	SVD-LLM	7.94	15.84	0.31	0.71	0.68	0.49	0.71	0.22	0.52	0.24	0.06
	SVD-LLM V2	7.12 (↓10%)	10.47 (↓34%)	0.32	0.72	0.70	0.52	0.75	0.24	0.54 (↑4%)	0.27 (+0.03)	0.07 (+0.01)
OPT-6.7B	Original	10.87	12.50	0.28	0.66	0.65	0.50	0.76	0.25	0.52	0.29	0.01
	FWSVD	14559	17898	0.03	0.08	0.02	0.01	0.05	0.01	0.03	0.01	0.00
	ASVD	82	102	0.16	0.41	0.30	0.36	0.61	0.07	0.32	0.09	0.00
	SVD-LLM	16.04	21.27	0.21	0.56	0.59	0.47	0.73	0.21	0.46	0.22	0.00
	SVD-LLM V2	13.46 (↓16%)	17.72 (↓17%)	0.25	0.61	0.62	0.49	0.74	0.22	0.49 (↑7%)	0.24 (+0.02)	0.01 (+0.01)
LLaMA-3 8B	Original	6.14	9.47	0.35	0.80	0.73	0.60	0.80	0.40	0.61	0.49	0.45
	FWSVD	4782	8195	0.01	0.04	0.01	0.02	0.02	0.01	0.02	0.00	0.00
	ASVD	17.55	28.41	0.20	0.59	0.61	0.41	0.69	0.30	0.47	0.37	0.28
	SVD-LLM	11.82	20.05	0.29	0.77	0.64	0.51	0.72	0.30	0.54	0.45	0.31
	SVD-LLM V2	8.01 (↓32%)	11.72 (↓42%)	0.33	0.79	0.70	0.58	0.77	0.36	0.59 (↑9%)	0.46 (+0.01)	0.40 (+0.09)

近似效果提升显著，性能损失大幅减少

SVDLLM vs. 剪枝 vs. 量化

	PERPLEXITY (↓) UNDER WEIGHT MEMORY BUDGET			
METHOD	10 GB	9 GB	8 GB	7 GB
LLM-Pruner	9.88	12.21	18.94	21.68
SliceGPT	8.78	12.73	16.39	27.41
BlockPruner	9.40	12.76	19.78	43.05
SVD-LLM V2	7.84 (↓17%)	8.48 (↓34%)	10.17 (↓49%)	15.62 (↓28%)

明显优于剪枝

METHOD	WEIGHT MEMORY	PERPLEXITY
GPTQ-3bit	2.8 GB	16.28
SVD-LLM V2	2.8 GB	119
SVD-LLM V2 + GPTQ-4bit	2.8 GB	9.97 (↓39%)

METHOD	DATA TYPE	WEIGHT MEMORY	PERPLEXITY
PB-LLM	1-bit	1.9 GB	104.83
BiLLM	1-bit	1.5 GB	47.67
SVD-LLM V2	16-bit	1.5 GB	99.64
SVD-LLM V2	2-bit	1.5 GB	14.73 (↓69%)

单独不如量化，但集成良好

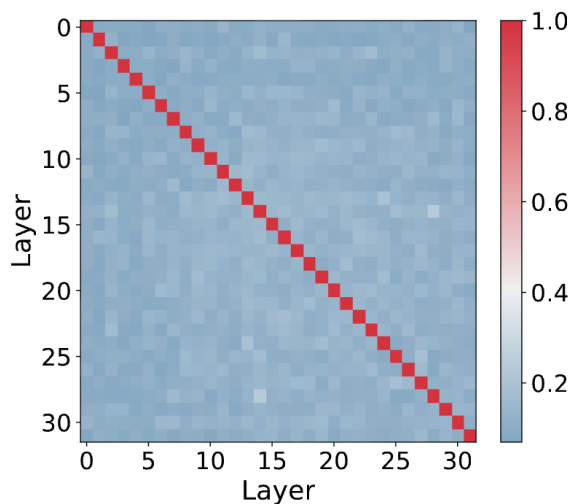
xKV: Cross-Layer SVD for KV-Cache Compression

Chi-Chih Chang¹ Chien-Yu Lin² Yash Akhauri¹ Wei-Cheng Lin³

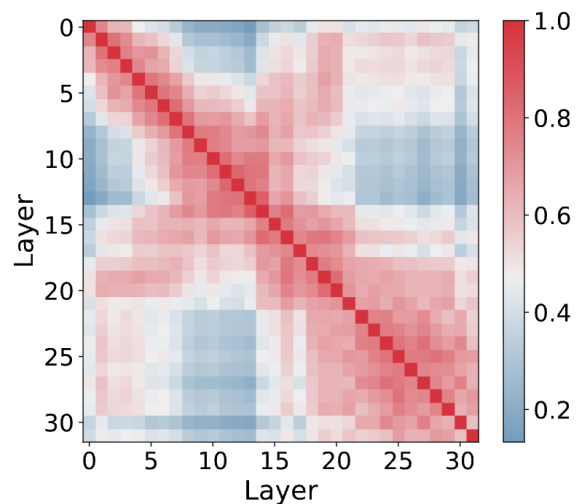
Kai-Chiang Wu³ Luis Ceze² Mohamed S. Abdelfattah¹

¹Cornell University ²University of Washington

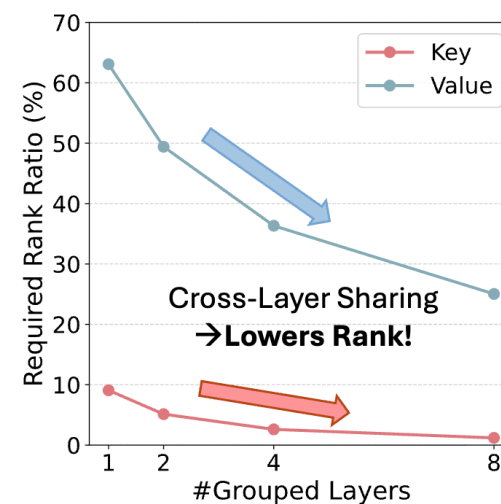
³National Yang Ming Chiao Tung University



跨层Value相似
度

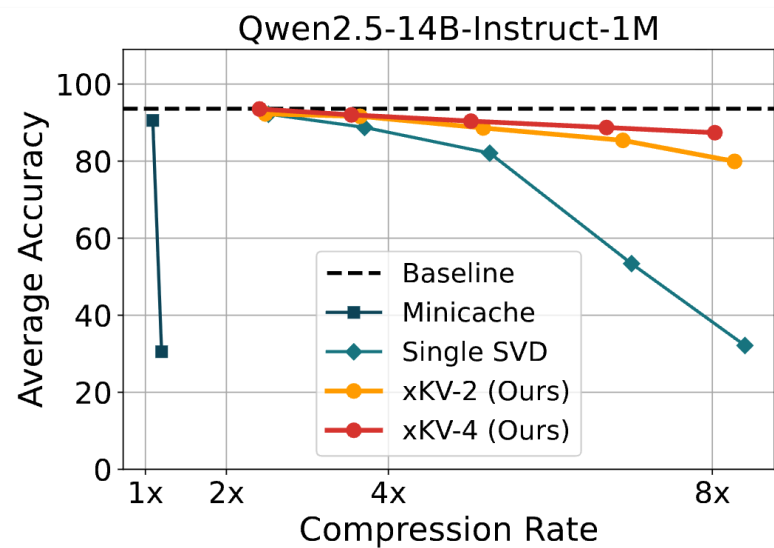
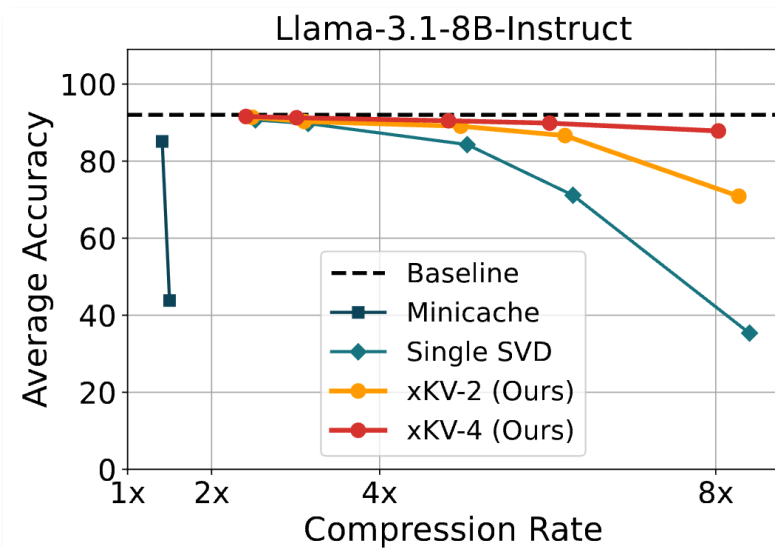
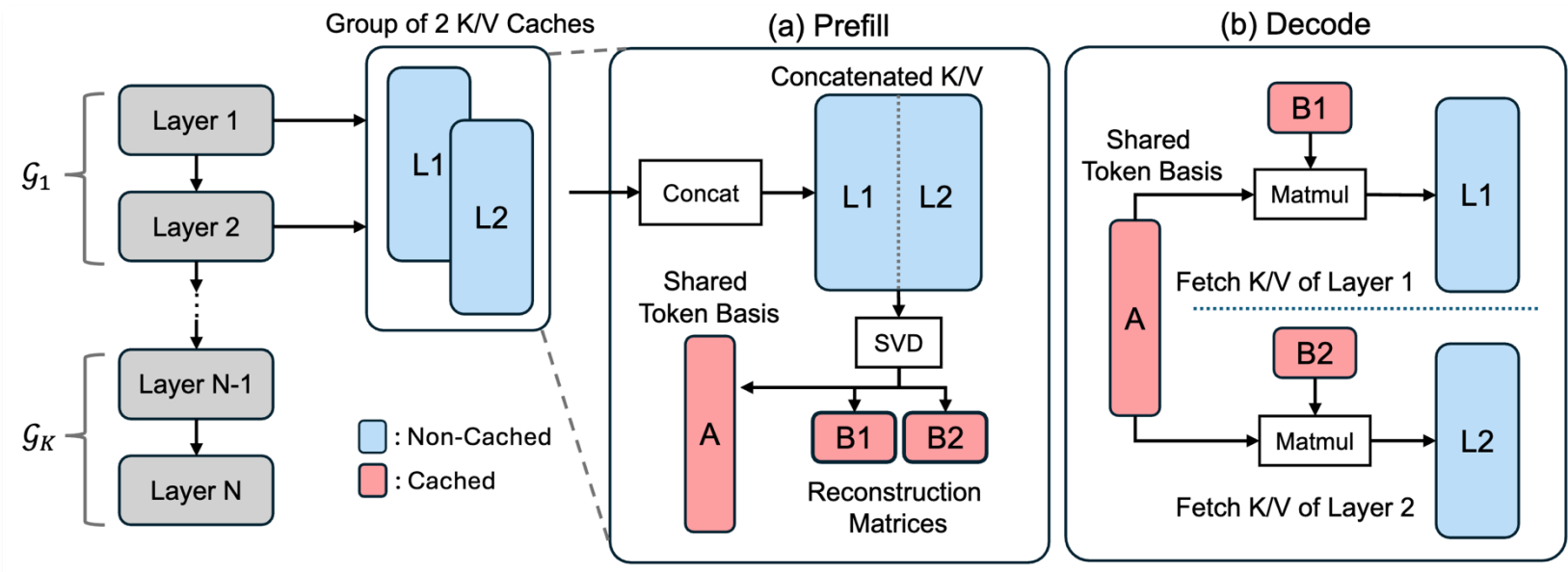


跨层Value低秩空间相似
度



95%性能最小所需
Rank

层共享低秩近似



层共享低秩近似

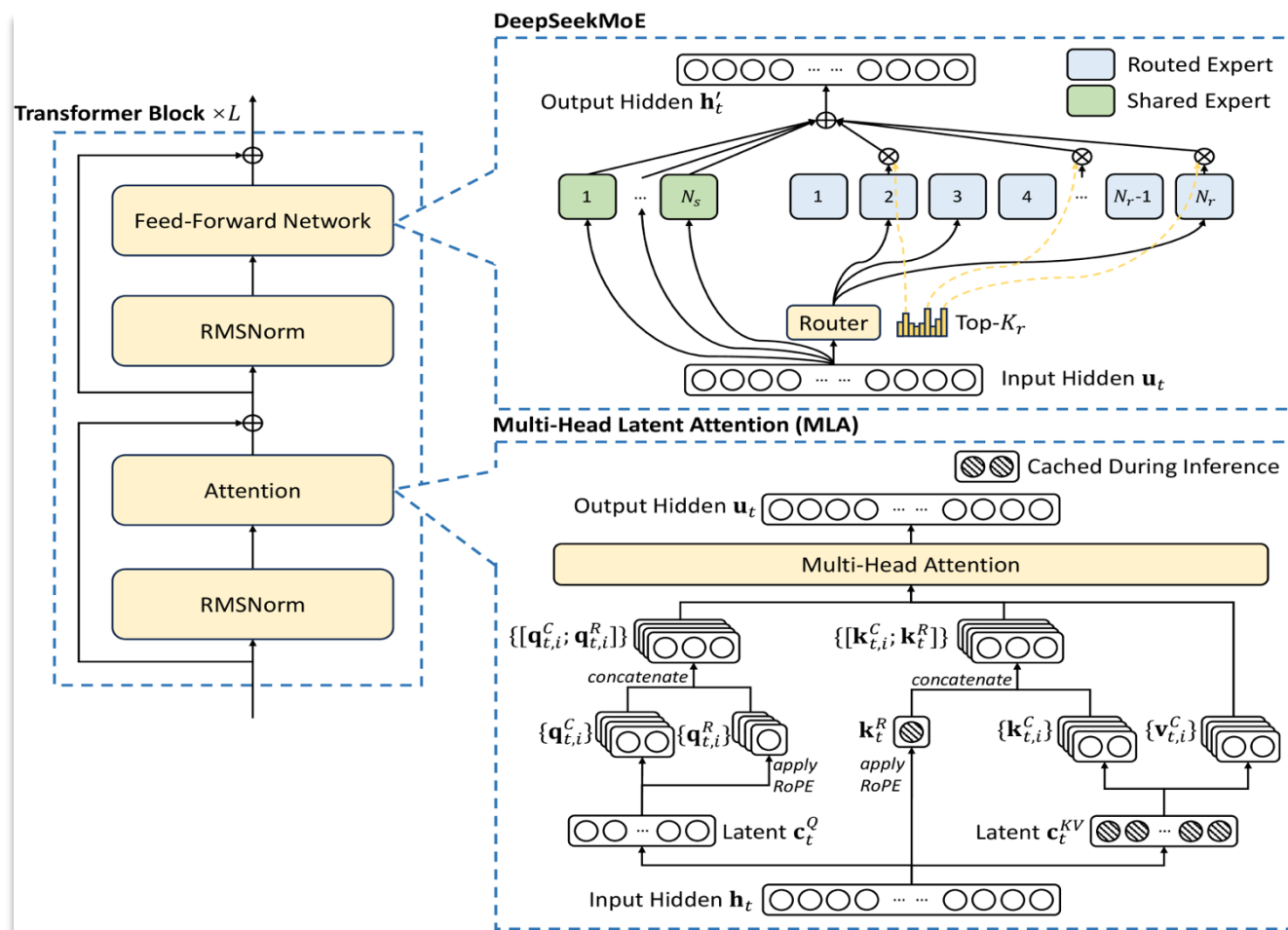
Method	Comp.	N-S1	N-S2	N-MK1	N-MK2	N-MQ	N-MV	QA-1	QA-2	VT	FWE	Avg.
Llama-3.1-8B-Instruct												
Baseline	1.0	100.0	100.0	99.0	97.9	99.0	98.4	83.3	60.4	97.3	84.7	92.0
Minicache	1.2	100.0	100.0	97.9	90.6	87.0	81.0	78.1	47.9	84.6	84.0	85.1
Minicache	1.3	87.5	64.6	39.6	10.4	13.3	20.1	60.4	35.4	49.0	58.0	43.8
Single SVD	2.5	100.0	100.0	100.0	97.9	97.9	96.1	80.2	58.3	96.9	79.5	90.7
xKV-2 (Ours)	2.5	100.0	100.0	100.0	97.9	97.9	96.3	83.3	61.5	96.2	80.6	91.4
xKV-4 (Ours)	2.4	100.0	100.0	100.0	97.9	98.4	97.1	84.4	60.4	96.2	81.2	91.6
Single SVD	8.4	29.2	26.0	32.3	96.9	8.6	17.2	44.8	36.5	2.7	59.0	35.3
xKV-2 (Ours)	8.3	99.0	75.0	69.8	97.9	74.5	65.9	67.7	49.0	36.9	73.3	70.9
xKV-4 (Ours)	8.0	100.0	96.9	97.9	97.9	95.3	93.5	76.0	54.2	87.7	78.8	87.8
Qwen2.5-7B-Instruct-1M												
Baseline	1.0	100.0	100.0	100.0	100.0	100.0	95.6	83.3	59.4	90.8	86.5	91.6
Minicache	1.2	78.1	34.4	32.3	3.1	27.3	42.7	26.0	24.0	8.5	8.0	28.4
Minicache	1.3	25.0	0.0	0.0	0.0	0.0	0.0	13.5	13.5	0.8	4.5	5.7
Single SVD	2.3	100.0	100.0	99.0	99.0	99.7	92.7	75.0	58.3	80.8	75.7	88.0
xKV-2 (Ours)	2.3	100.0	100.0	100.0	99.0	100.0	90.6	80.2	59.4	79.0	81.2	88.9
xKV-4 (Ours)	2.2	100.0	100.0	100.0	99.0	100.0	90.9	83.3	60.4	85.0	83.0	90.2
Single SVD	6.4	24.0	7.3	6.2	97.9	4.4	3.4	36.5	35.4	6.9	26.4	24.8
xKV-2 (Ours)	6.3	99.0	81.2	75.0	97.9	38.3	57.0	56.2	46.9	46.7	41.3	64.0
xKV-4 (Ours)	6.2	100.0	85.4	91.7	99.0	84.1	76.8	62.5	50.0	66.2	64.9	78.1
Qwen2.5-14B-Instruct-1M												
Baseline	1.0	100.0	100.0	100.0	99.0	100.0	99.2	80.2	66.7	99.6	91.3	93.6
Minicache	1.1	100.0	100.0	100.0	99.0	98.2	99.0	72.9	61.5	90.8	84.4	90.6
Minicache	1.3	0.0	1.0	0.0	0.0	0.0	0.0	17.7	27.1	0.2	8.3	5.4
Single SVD	2.5	100.0	100.0	100.0	99.0	100.0	96.6	78.1	62.5	98.5	87.2	92.2
xKV-2 (Ours)	2.5	100.0	100.0	100.0	99.0	100.0	95.6	81.2	61.5	99.2	86.1	92.2
xKV-4 (Ours)	2.4	100.0	100.0	100.0	99.0	100.0	95.3	83.3	69.8	99.4	88.5	93.5
Single SVD	8.4	12.5	9.4	18.8	96.9	25.5	14.3	32.3	44.8	8.1	59.0	32.2
xKV-2 (Ours)	8.3	95.8	91.7	91.7	96.9	90.4	74.0	49.0	52.1	77.7	80.2	79.9
xKV-4 (Ours)	8.0	100.0	96.9	99.0	97.9	97.1	88.5	63.5	58.3	86.0	86.5	87.4

4层共享>2层共享

性能损失较小

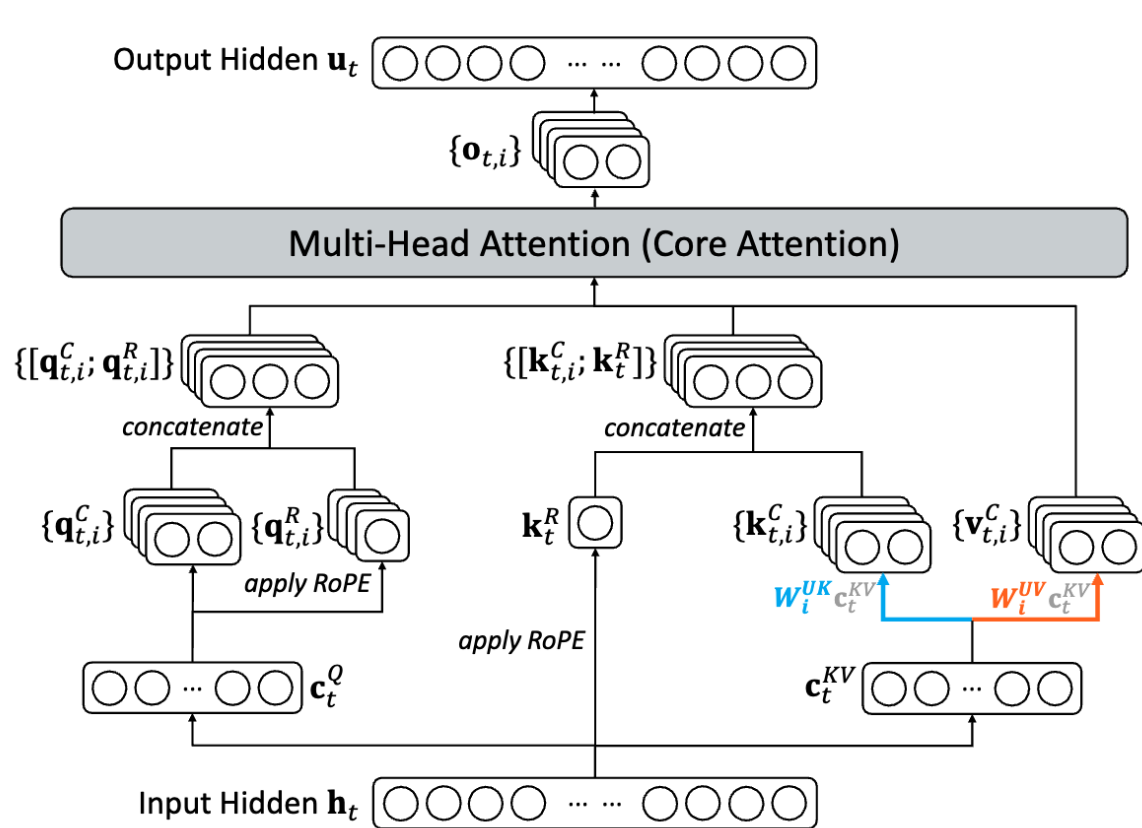
DeepSeek-MLA本质上是寻找低秩子空间

DeepSeek提出的Multi-head Latent Attention (MLA) 660/865
TFlops@H800



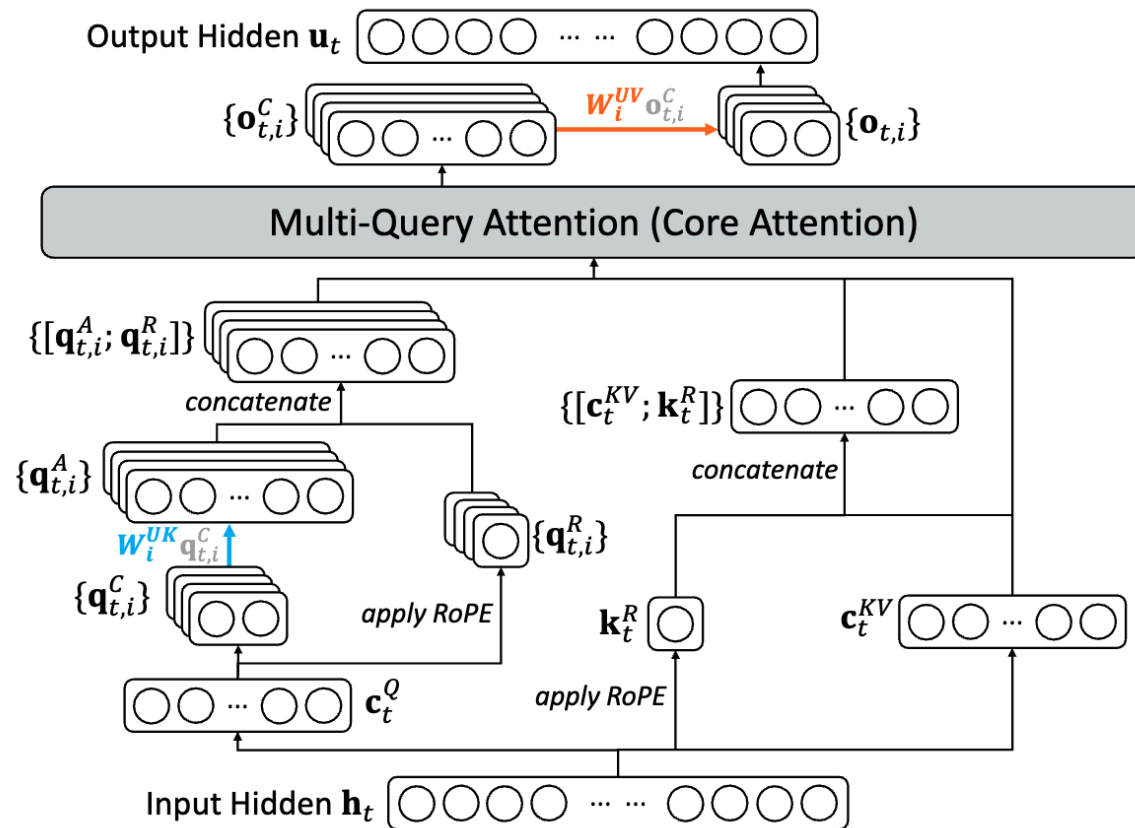
MLA被视为新架构
需从头大规模预训练

2 mode of MLA



(a) MHA mode of MLA.

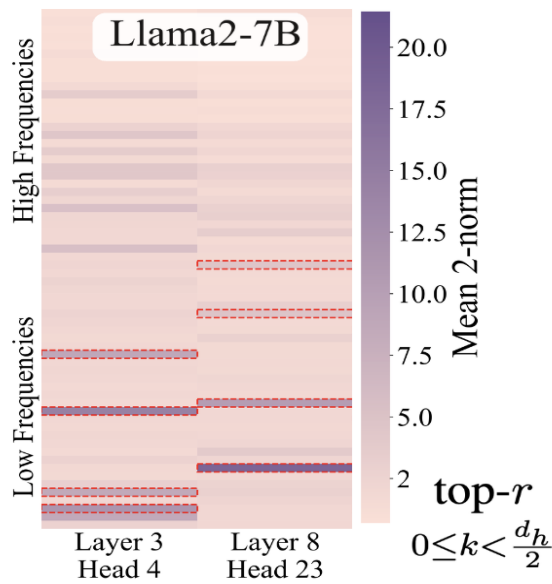
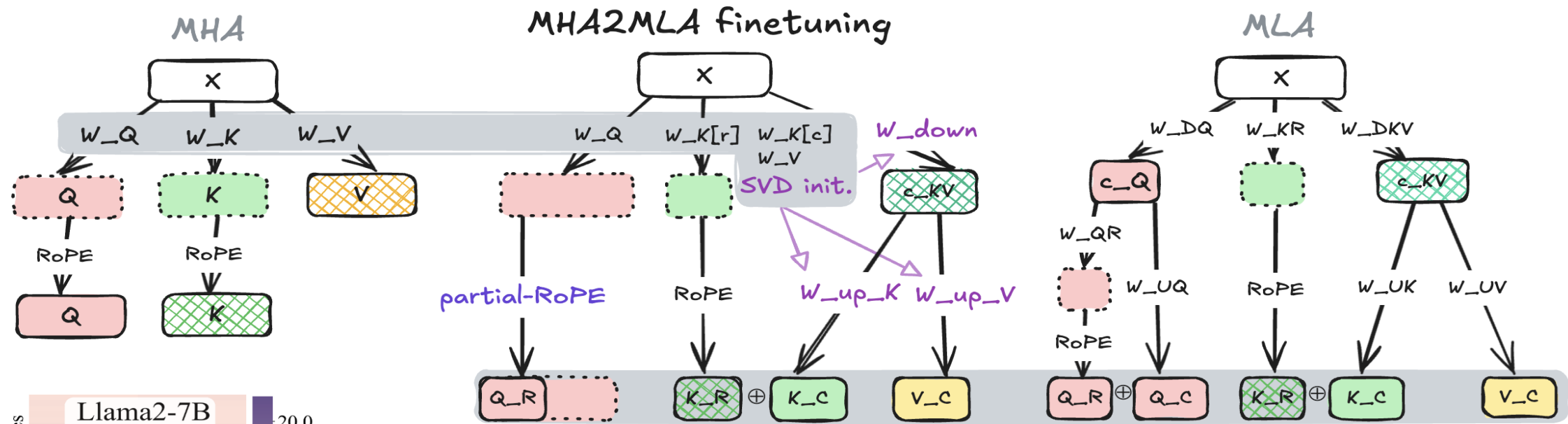
训练+Prefilling



(b) MQA mode of MLA.

Decoding

MHA2MLA

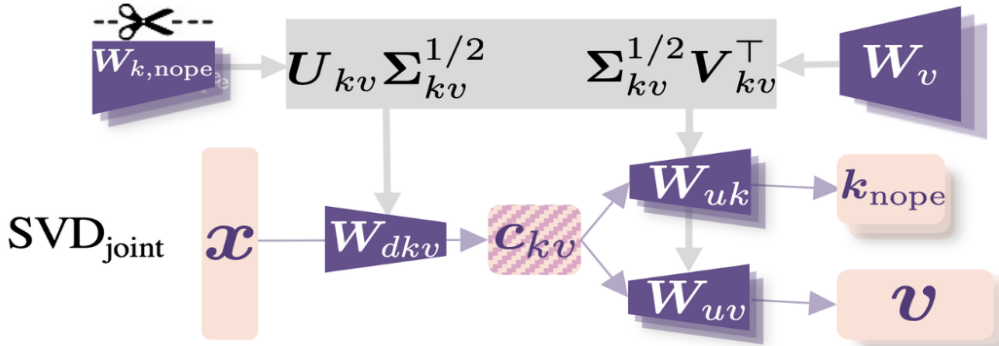


① Partial-RoPE

$$q_i = \begin{bmatrix} q_i^{[2k, 2k+1]} \end{bmatrix}_{0 \leq k < \frac{d_h}{2}}$$
$$k_i = \begin{bmatrix} k_i^{[2k, 2k+1]} \end{bmatrix}_{0 \leq k < \frac{d_h}{2}}$$

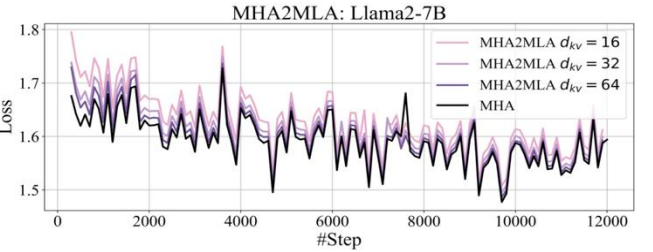
$$\left(\left\| q_*^{[2k, 2k+1]} \right\| \left\| k_*^{[2k, 2k+1]} \right\| \right)$$

② Low-rank SVD



MHA2MLA

Model		Tokens	KV Mem.	Avg.
135MSmolLM		600B		44.50
- GQA	$d_{kv} = 128$			44.42
	$d_{kv} = 32$	6B	-68.75%	43.21 ^{-1.21}
- GQA2MLA	$d_{kv} = 16$	(1%)	-81.25%	42.18 ^{-2.24}
	$d_{kv} = 8$		-87.50%	41.04 ^{-3.38}
360MSmolLM		600B		49.60
- GQA	$d_{kv} = 128$			49.51
	$d_{kv} = 32$	6B	-68.75%	48.14 ^{-1.37}
- GQA2MLA	$d_{kv} = 16$	(1%)	-81.25%	46.88 ^{-2.63}
	$d_{kv} = 8$		-87.50%	45.84 ^{-3.67}
1B7SmolLM		1T		55.90
- MHA	$d_{kv} = 128$			55.71
	$d_{kv} = 32$	6B	-68.75%	54.66 ^{-1.05}
- MHA2MLA	$d_{kv} = 16$	(0.6%)	-81.25%	54.28 ^{-1.43}
	$d_{kv} = 8$		-87.50%	52.79 ^{-2.92}
7BLlama2		2T		59.85
- MHA	$d_{kv} = 256$			59.50
	$d_{kv} = 64$	12B	-68.75%	59.21 ^{-0.29}
- MHA2MLA	$d_{kv} = 32$	(0.6%)	-81.25%	59.20 ^{-0.30}
	$d_{kv} = 16$		-87.50%	58.48 ^{-1.02}
13BLlama2		2T		62.65
- MHA	$d_{kv} = 256$			60.72
	$d_{kv} = 64$	12B	-68.75%	60.45 ^{-0.27}
- MHA2MLA	$d_{kv} = 32$	(0.6%)	-81.25%	60.49 ^{-0.23}
	$d_{kv} = 16$		-87.50%	59.68 ^{-1.04}



MHA2MLA Public

Towards Economical Inference: Enabling DeepSeek's Multi-Head Latent Attention in Any Transformer-based LLMs

Python 177 21

alphaXiv @askalphaxiv · 2月24日

The efficiency breakthrough in DeepSeek's R1 architecture can now be ported to other LLMs!

New method needs just 0.3~0.6% of original training data:

- Slashes KV cache by 92%
- Minimal performance impact
- Works with Llama2-7B 🔥

Trending on alphaXiv 🚀



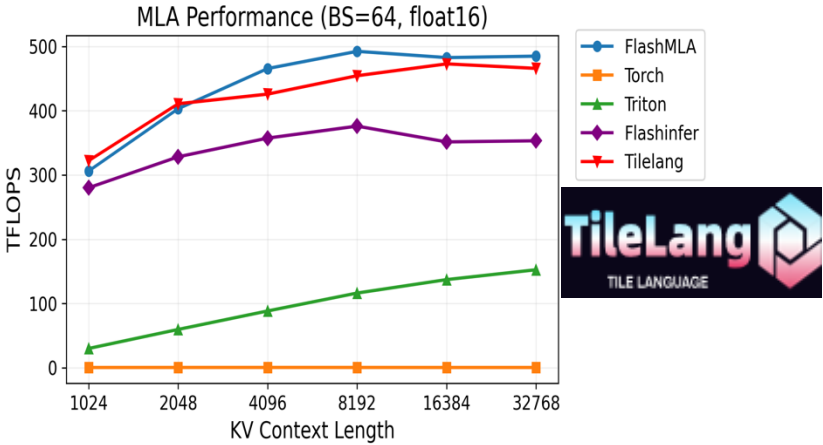
7 107 694 6.7万

alphaXiv @askalphaxiv · 2月24日

Towards Economical Inference: Enabling DeepSeek's Multi-Head Latent Attention in Any Transformer-based LLMs

See the paper here: alphaxiv.org/abs/2502.14837

Code: github.com/JT-Ushio/MHA2M...



LLaMA2-7B(d_kv_16)	Bsz	SeqLen	Mem. [MB]	Attention's Latency [ms]
MHA w. FlashAttention2	8	2K	29743	5.95
	8	4K	29671	10.66
	64	2K	OOM	43.74
MLA w. Triton Kernel	8	2K	15128(-49%)	2.94 (-51%)
	8	4K	21273(-28%)	3.90 (-63%)
	64	2K	24932(/)	15.17 (-65%)
MLA w. TileLang Kernel	8	2K	15128(-49%)	3.52 (-41%)
	8	4K	21273(-28%)	6.08 (-43%)
	64	2K	24932(/)	4.10 (-91%)

$$\begin{aligned} Q \cdot K^\top &= (X \cdot W_q) \cdot (X \cdot W_{dk} \cdot W_{uk})^\top \\ &= (X \cdot W_q \cdot W_{uk}^\top) \cdot (X \cdot W_{dk})^\top \\ &= (Q \cdot W_{uk}^\top) \cdot C_k^\top \end{aligned}$$

$$\begin{aligned} Q \cdot K^\top &= \text{RMS-Norm}(X \cdot W_q) \cdot \text{RMS-Norm}(X \cdot W_{dk} \cdot W_{uk})^\top \\ &= Q \cdot \text{RMS-Norm}(C_k \cdot W_{uk})^\top \\ &= Q \cdot \frac{(C_k \cdot W_{uk})^\top \cdot \gamma_k}{\|(C_k \cdot W_{uk})^\top\|_{rms}} \\ &= Q \cdot \frac{(C_k \cdot W'_{uk})^\top}{\|C_k^\top\|_{rms} \cdot \mathbb{E}\left(\frac{\|(C_k \cdot W_{uk})^\top\|_{rms}}{\|C_k^\top\|_{rms}}\right)} \\ &= Q \cdot W'_{uk}{}^\top \cdot \text{RMS-Norm}(C_k^\top) \cdot \mathbb{E}\left(\frac{\|(C_k \cdot W_{uk})^\top\|_{rms}}{\|C_k^\top\|_{rms}}\right) \end{aligned}$$

适配Qwen3 QKNorm

Algorithm 1: Pseudocode of Modality-Decoupled SVD

Input: W — Joint KV weight matrix
 $\mathbf{X}_{visual}, \mathbf{X}_{text}$ — Visual / Text activations
 r_{visual}, r_{text} — Target ranks
Output: $\{\mathbf{W}_m^{up}, \mathbf{W}_m^{down}\}_{m \in \{visual, text\}}$

- 1: **procedure** MD_SVD_COMPRESS($W, \mathbf{X}_{visual}, \mathbf{X}_{text}, r_{visual}, r_{text}$)
- 2: **for** $m \in \{Visual, Text\}$ **do** ▷ process each modality
- 3: $\mathbf{S}_m \leftarrow \mathbf{X}_m \mathbf{X}_m^\top$
- 4: $[\mathbf{U}_s, \Sigma_s, _] \leftarrow \text{SVD}(\mathbf{S}_m)$
- 5: $\mathbf{D} \leftarrow W \mathbf{U}_s \Sigma_s^{1/2}$
- 6: $[\mathbf{U}_d, \Sigma_d, \mathbf{V}_d] \leftarrow \text{SVD}(\mathbf{D})$
- 7: Keep first r_m components of $(\mathbf{U}_d, \Sigma_d, \mathbf{V}_d)$
- 8: $\mathbf{W}_m^{up} \leftarrow \mathbf{U}_d \Sigma_d^{1/2}$
- 9: $\mathbf{W}_m^{down} \leftarrow \Sigma_d^{1/2} \mathbf{V}_d \Sigma_s^{-1/2} \mathbf{U}_s^{-1}$
- 10: **end for**
- 11: **return** $\{\mathbf{W}_m^{up}, \mathbf{W}_m^{down}\}_m$
- 12: **end procedure**

适配视觉语言模型VLM

MHA2MLA

Model		Tokens	KV Mem.	Avg.	AI2D	GQA	POPE	SEED ^I	RWQ	MMB ^{EN}	Chart	Doc
LLaVA-1.5 _{7B}				63.89	53.63	61.27	84.03	65.28	56.21	62.89	-	-
- MHA	$d_{kv} = 256$	0.5B		64.13	54.92	61.97	83.96	64.80	56.34	62.80	-	-
	$d_{kv} = 64$		-62.50%	63.84 _{-0.29}	54.05	62.32	82.74	65.17	54.51	64.26	-	-
- MHA2MLA	$d_{kv} = 32$		-75.00%	63.58 _{-0.55}	52.49	62.06	83.18	65.04	54.90	63.83	-	-
	$d_{kv} = 16$		-81.30%	62.27 _{-1.86}	48.41	62.08	83.61	65.85	51.37	62.29	-	-
LLaVA-NeXT _{8B}				70.72	70.56	65.13	87.18	72.46	58.69	71.82	68.44	71.47
- GQA	$d_{kv} = 256$	1.8B		70.78	70.76	65.30	87.22	71.95	59.22	71.74	69.04	70.99
	$d_{kv} = 128$		-84.38%	70.23 _{-0.55}	69.53	65.25	85.96	71.90	59.08	72.08	68.68	69.34
- GQA2MLA	$d_{kv} = 64$		-90.63%	68.75 _{-2.03}	68.20	64.45	86.32	71.57	58.56	69.50	65.56	65.83
	$d_{kv} = 32$		-93.75%	66.72 _{-4.06}	65.90	63.98	86.56	71.15	56.86	63.32	63.60	62.41
Qwen2.5-VL _{7B}				79.47	82.58	60.42	86.22	77.34	68.37	82.82	83.24	94.78
- MHA	$d_{kv} = 256$	0.5B		80.75	83.35	63.26	87.50	76.83	69.67	84.28	86.88	94.29
	$d_{kv} = 128$		-90.36%	80.63 _{-0.12}	82.71	63.26	87.54	76.83	69.41	84.02	86.80	94.48
- MHA2MLA	$d_{kv} = 64$		-94.64%	79.47 _{-1.28}	80.63	63.08	87.85	76.51	68.76	81.35	84.76	92.81
	$d_{kv} = 32$		-96.43%	77.02 _{-3.73}	78.11	62.87	87.42	74.75	66.54	77.15	82.16	87.17

同样适用于多模态大模型，且视觉模态压缩空间更大

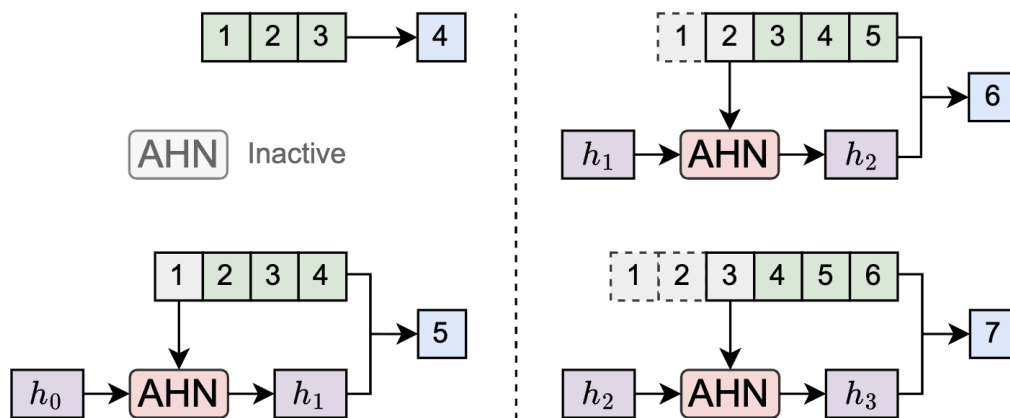
MHA2Linear Attention

Artificial Hippocampus Networks for Efficient Long-Context Modeling

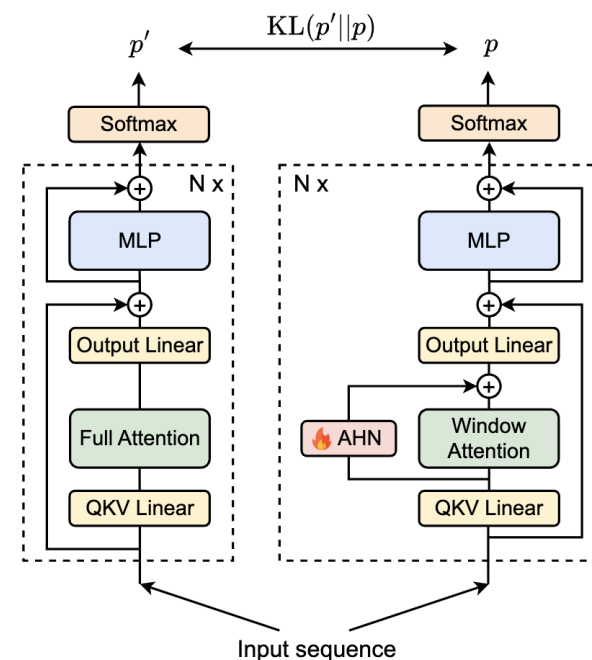
Yunhao Fang*, Weihao Yu^{*,†}, Shu Zhong, Qinghao Ye, Xuehan Xiong[¶], Lai Wei

ByteDance Seed

Window token Token to be compressed Evicted token Compressed memory Generated token

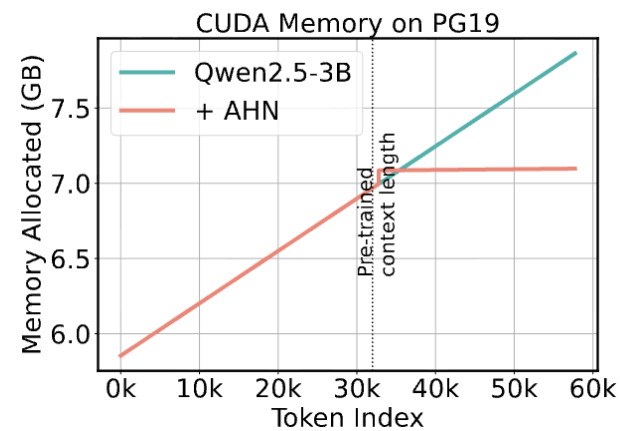
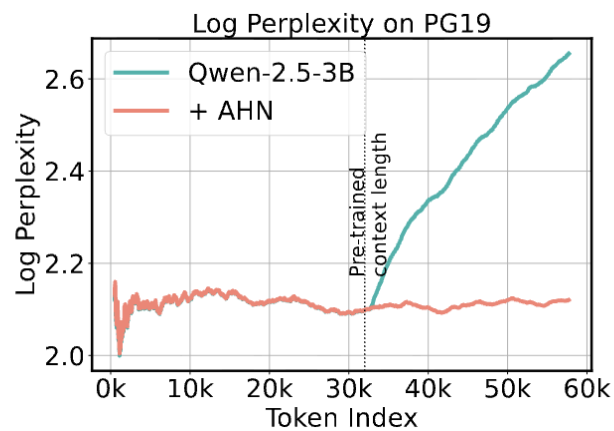
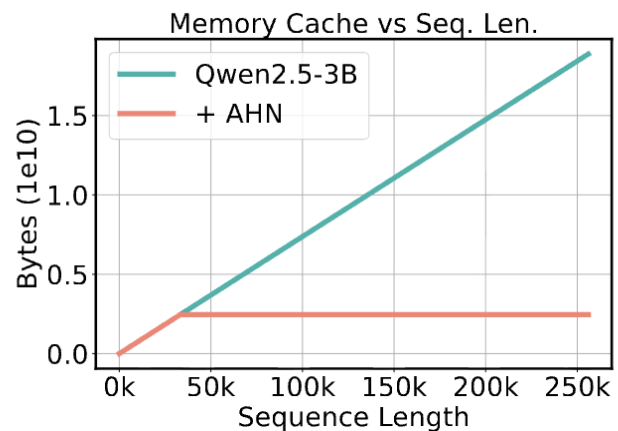
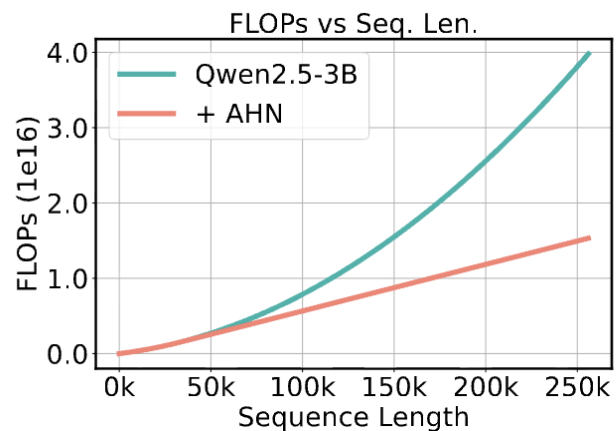


(a)



(b)

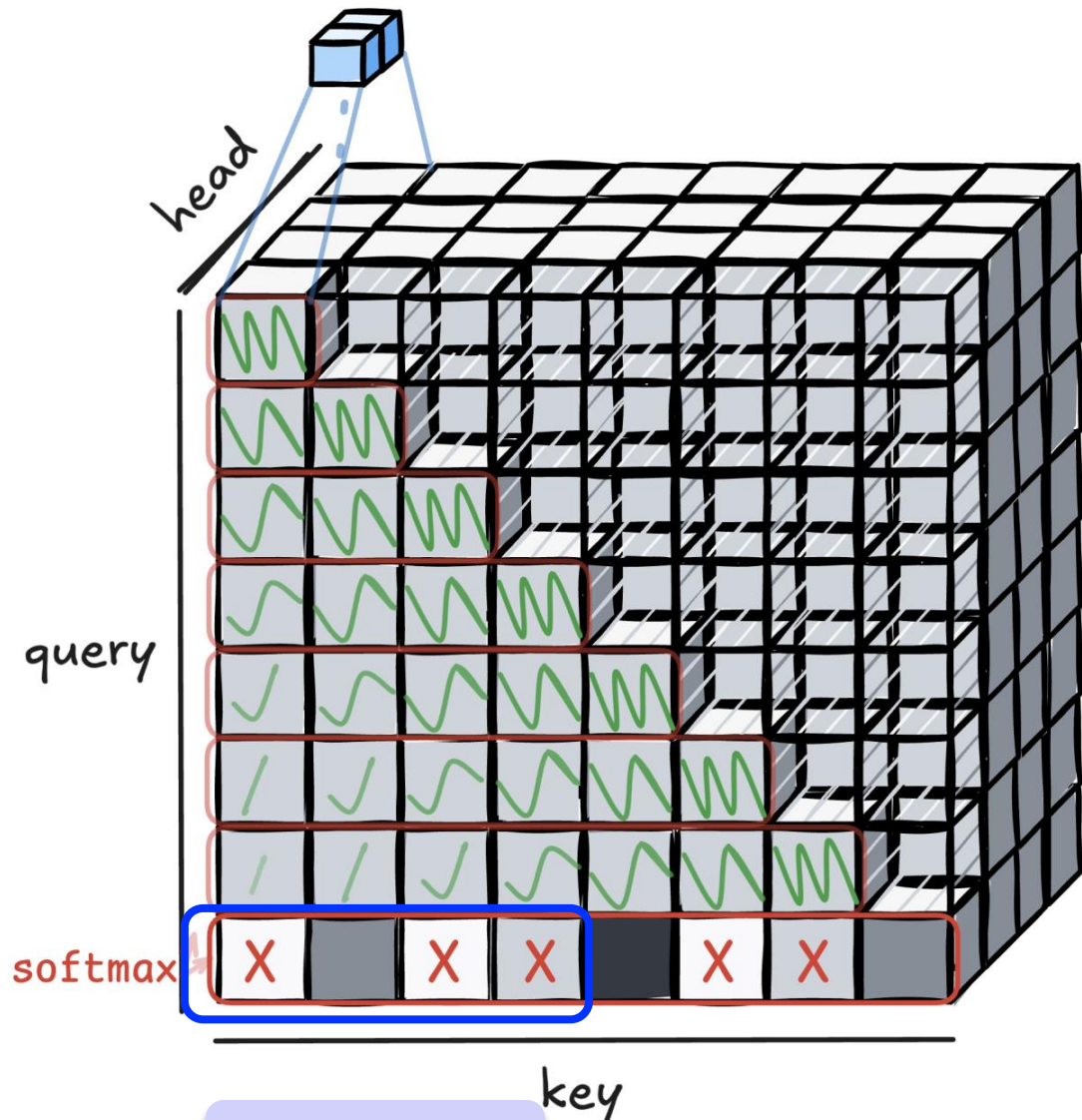
MHA2Linear Attention



MHA2Linear Attention

Base model	Token mixer	Extra param ratio	Mixing FLOP ratio	Model FLOP ratio	Memory cache ratio	LV-Eval				InfiniteBench		
						cmrc -mixup	loogle-SD -mixup	dureader -mixup	Avg.*	En. QA	Zh. QA	Avg.
Qwen2.5-3B-Instruct	Full Attn	0%	100%	100%	100%	7.28	0.89	13.22	4.41	7.28	11.75	9.52
	Sinks + SWA	0%	46.6%	59.3%	25.6%	7.48	4.59	11.49	4.59	8.63	12.31	10.47
	CT-Max	0%	47.1%	59.7%	26.0%	6.10	3.88	11.37	4.12	7.40	12.59	10.00
	CT-Average	0%	47.1%	59.7%	26.0%	6.95	4.70	11.40	4.47	8.30	13.32	10.81
	AHN-Mamba2	0.4%	46.7%	59.4%	26.0%	7.84	5.20	12.35	5.13	9.29	15.58	12.44
	AHN-DN	0.4%	46.7%	59.4%	26.0%	9.41	<u>5.99</u>	11.49	<u>5.68</u>	10.61	16.41	13.51
	AHN-GDN	0.4%	46.7%	59.4%	26.0%	<u>7.96</u>	7.21	<u>12.52</u>	5.88	10.61	<u>15.87</u>	<u>13.24</u>
Qwen2.5-7B-Instruct	Full Attn	0%	100%	100%	100%	4.30	0.17	12.8	3.62	11.23	15.76	13.50
	Sinks + SWA	0%	48.0%	65.2%	25.6%	9.52	4.76	14.09	5.34	10.66	15.66	13.16
	CT-Max	0%	48.5%	65.6%	26.0%	8.35	4.02	12.34	4.82	10.56	15.45	13.00
	CT-Average	0%	48.5%	65.6%	26.0%	9.48	4.86	13.78	5.28	10.63	15.99	13.31
	AHN-Mamba2	0.2%	48.1%	65.4%	26.0%	<u>12.57</u>	<u>5.54</u>	14.13	6.21	11.36	17.06	14.21
	AHN-DN	0.2%	48.1%	65.4%	26.0%	11.97	5.67	16.52	6.82	<u>12.86</u>	<u>20.10</u>	<u>16.48</u>
	AHN-GDN	0.3%	48.1%	65.4%	26.0%	12.69	4.71	<u>15.30</u>	<u>6.54</u>	13.37	20.48	16.93
Qwen2.5-14B-Instruct	Full Attn	0%	100%	100%	100%	8.79	1.45	13.84	4.99	11.23	13.19	12.21
	Sinks + SWA	0%	49.5%	62.3%	25.6%	11.96	7.59	12.23	5.69	11.62	13.45	12.54
	CT-Max	0%	49.8%	62.6%	25.9%	10.55	7.53	12.08	5.28	10.58	12.73	11.66
	CT-Average	0%	49.8%	62.6%	25.9%	11.89	7.41	12.46	5.64	10.61	13.28	11.95
	AHN-Mamba2	0.3%	49.7%	62.5%	25.9%	<u>14.03</u>	7.20	15.39	6.43	14.21	16.20	15.21
	AHN-DN	0.3%	49.7%	62.5%	25.9%	13.13	9.14	<u>14.46</u>	<u>6.50</u>	16.54	<u>18.42</u>	17.48
	AHN-GDN	0.4%	49.7%	62.5%	25.9%	14.16	<u>8.54</u>	13.94	6.51	<u>14.48</u>	18.55	<u>16.52</u>

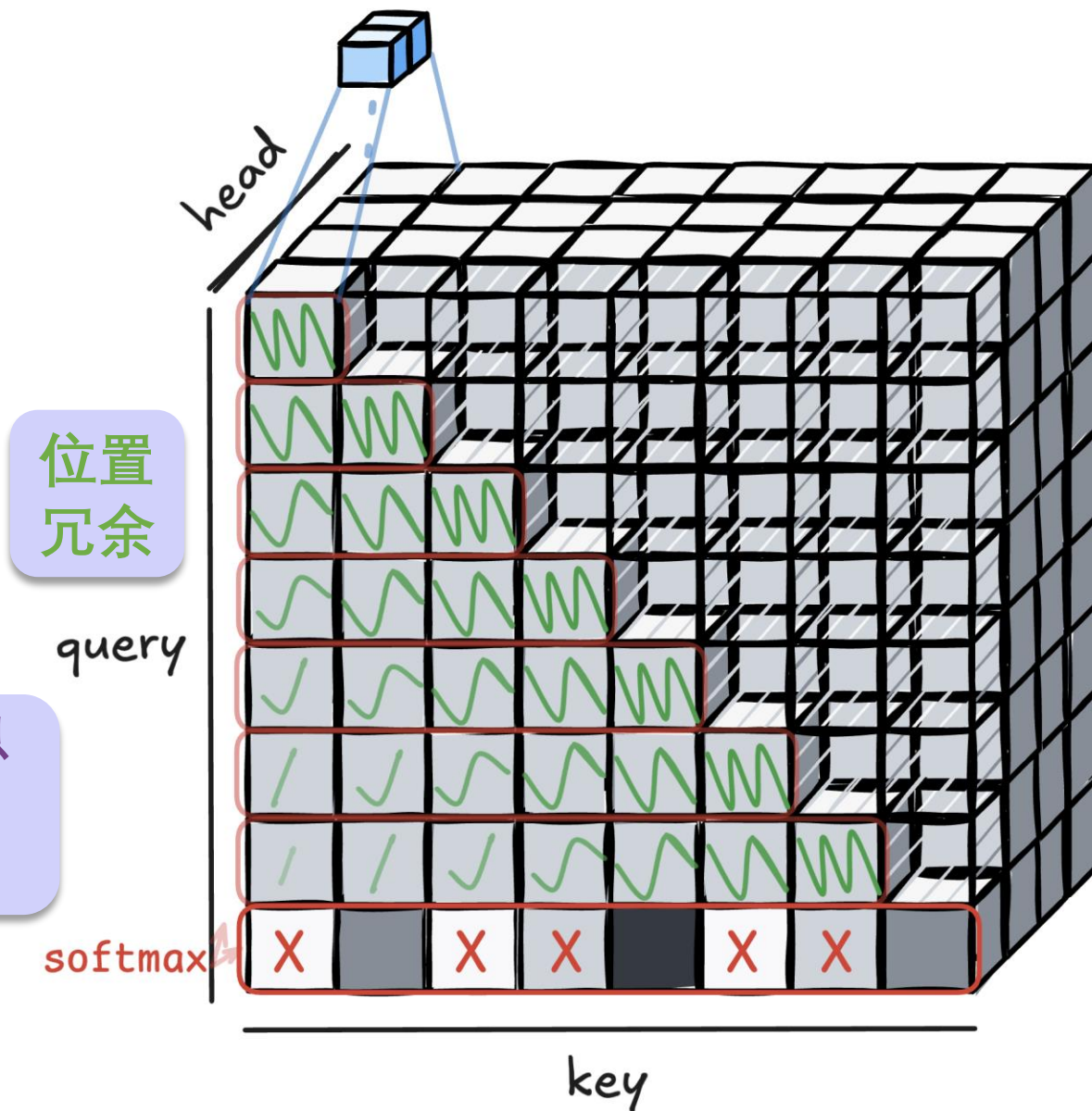
Attn内生的低秩性



Linear Transformation可被分解

低秩性

Attn内生的位置冗余性

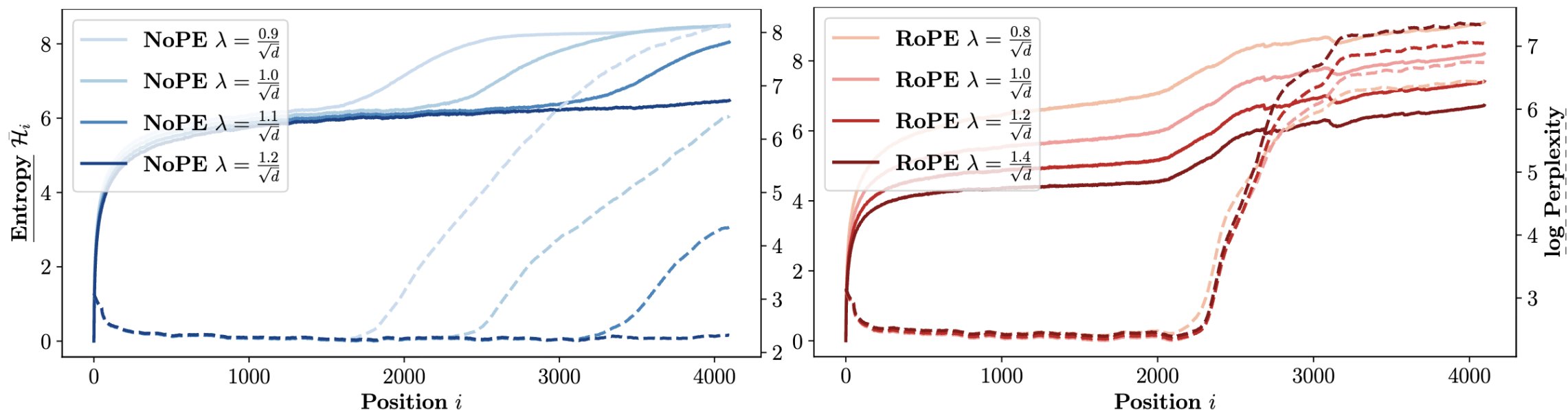


不需要精确的全位置信息
→更容易泛化
→更容易记忆

Attn内生的位置冗余性

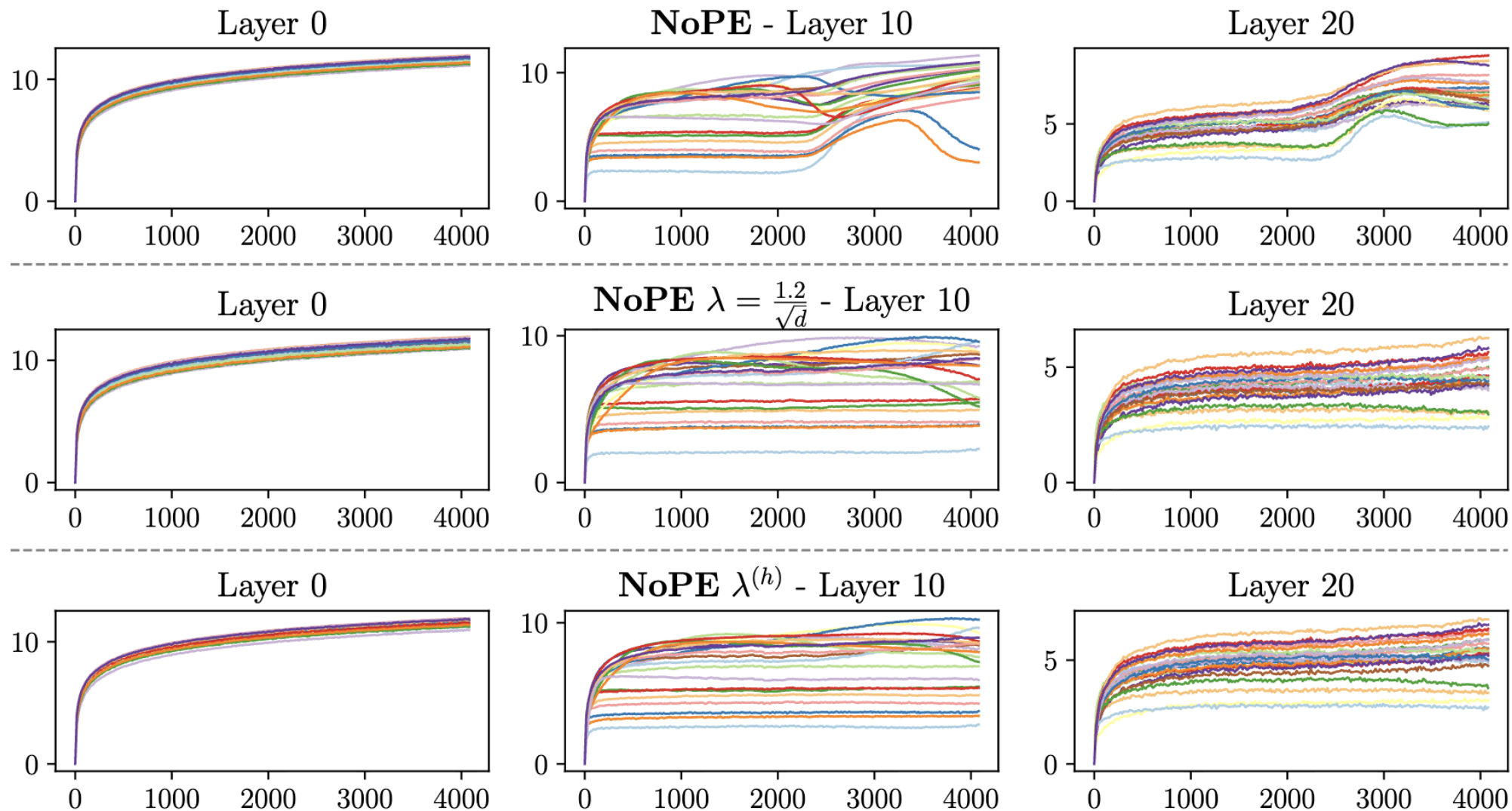
Length Generalization of Causal Transformers without Position Encoding

Jie Wang^{1*}, Tao Ji^{2*}, Yuanbin Wu¹,
Hang Yan⁵, Tao Gui³, Qi Zhang², Xuanjing Huang^{2,4}, Xiaoling Wang¹

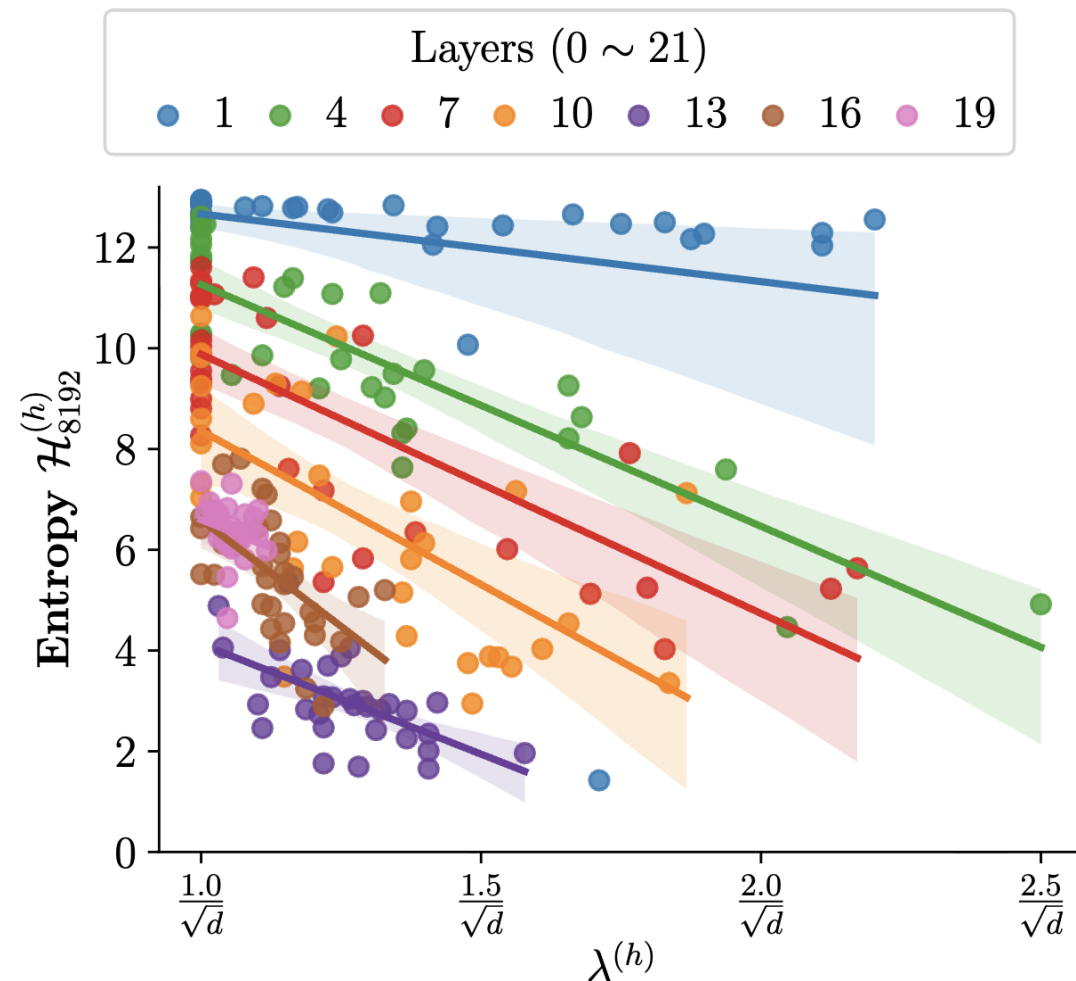
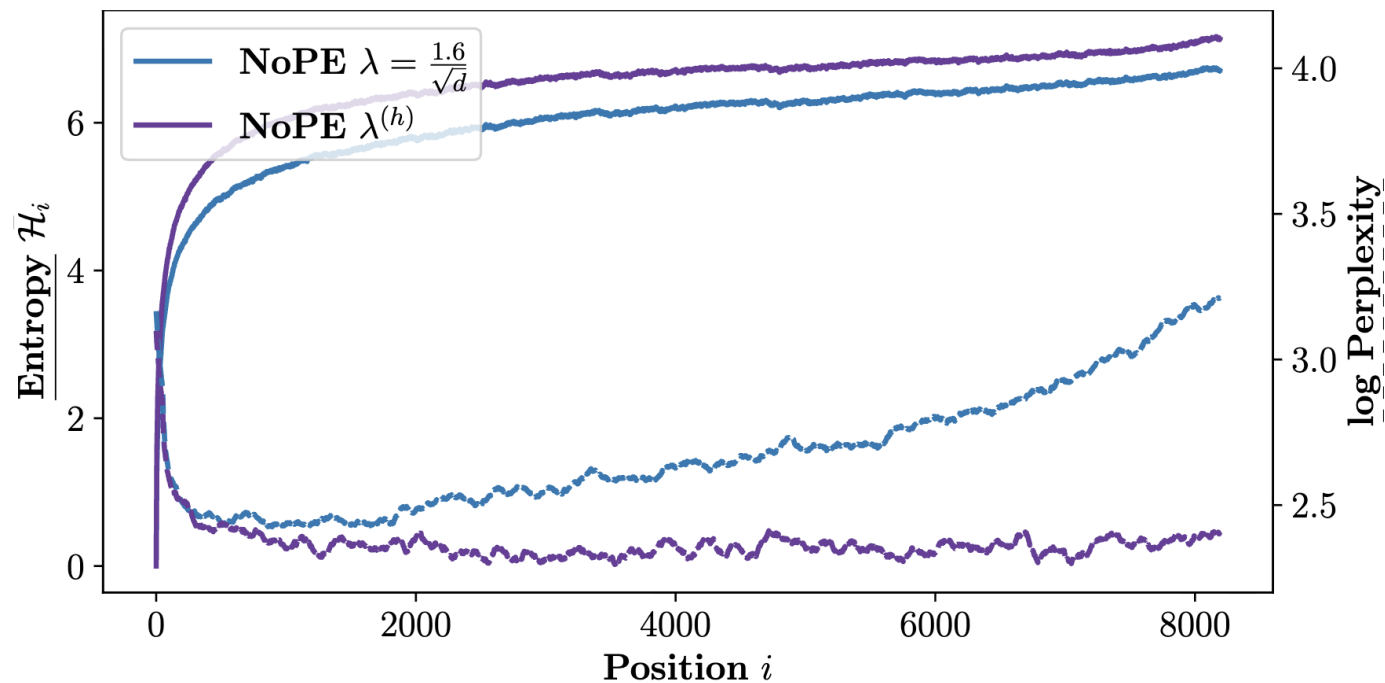


熵增代表着泛化失败

HeadScale



HeadScale



不同头的熵增表现不同

HeadScale

Model	Avg.	arc_challenge	arc_easy	boolq	hellaswag	openbookqa	piqa	winogrande
RoPE	46.1	24.3	44.9	59.7	43.5	29.8	67.3	53.3
NoPE	46.2	24.0	44.9	58.1	43.4	31.8	68.4	52.9

NoPE works

Table 1: Commonsense reasoning ability of the pre-trained base models.

Model	FT		PG19				Proof-pile			
	L'	Tokens	2K	4K	8K	16K	2K	4K	8K	16K
<i>Original LMs</i>										
RoPE	-	-	14.5	491.4	488.5	599.5	3.5	303.0	432.1	759.5
NoPE	-	-	14.6	326.9	$> 10^3$	$> 10^3$	3.5	117.4	$> 10^3$	$> 10^3$
BLOOM	-	-	27.7	158.0	264.6	403.4	6.9	74.1	176.2	334.5
MPT	-	-	10.6	103.6	361.6	345.1	2.8	70.1	$> 10^3$	$> 10^3$
<i>Generalization for RoPE</i>										
NTK ^{zero}	-	-	14.5	14.9	22.8	80.4	3.5	3.3	4.1	13.3
YaRN ^{zero}	-	-	14.5	14.5	15.0	17.1	3.5	3.3	3.2	3.6
PI ^{fair}	4K	6M	16.0	15.9	551.9	$> 10^3$	3.8	3.4	307.9	633.8
	8K	13M	17.4	17.1	17.1	752.8	4.0	3.6	3.4	406.3
	16K	30M	18.7	18.4	18.3	18.2	4.3	3.9	3.6	3.6
YaRN ^{fair}	4K	6M	15.5	15.4	545.2	$> 10^3$	3.7	3.4	351.5	698.2
	8K	13M	15.7	15.4	15.5	794.6	3.8	3.4	3.2	492.8
	16K	30M	15.9	15.6	15.4	15.5	3.8	3.5	3.2	3.2
PI ^{raw}	4K	33M	15.2	15.0	623.8	951.7	3.6	3.3	334.4	595.5
	8K	66M	15.4	15.1	15.0	909.6	3.6	3.3	3.0	463.0
	16K	131M	15.6	15.3	15.0	14.9	3.7	3.3	3.0	3.0
YaRN ^{raw}	4K	33M	15.1	15.0	573.3	951.4	3.6	3.3	358.8	656.8
	8K	66M	15.1	14.8	14.8	816.0	3.6	3.3	3.1	501.5
	16K	131M	15.0	14.8	14.5	14.5	3.6	3.3	3.0	3.0
<i>Generalization for NoPE</i>										
$\lambda = \frac{1.2}{\sqrt{d}}$	-	-	15.0	16.0	513.7	$> 10^3$	3.6	3.3	175.3	$> 10^3$
$\lambda = \frac{1.5}{\sqrt{d}}$	-	-	19.0	20.2	45.3	224.1	3.9	3.7	4.9	99.2
$\lambda = \frac{1.8}{\sqrt{d}}$	-	-	30.4	42.4	69.1	198.8	5.1	5.6	8.5	38.2
$\lambda^{(h)}$	4K	6M	14.8	15.3	404.5	$> 10^3$	3.5	3.2	153.4	$> 10^3$
	8K	13M	15.7	15.3	21.1	721.7	3.6	3.3	3.2	318.5
	18K	30M	18.3	19.0	18.8	30.4	4.0	3.7	3.3	4.1

HeadScale外推8
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innovation in the Llama 4 architecture is the use of interleaved attention layers without positional embeddings. Additionally, we employ inference time temperature scaling of attention to enhance length generalization. We call this the iRoPE architecture, where “i” stands for “interleaved” attention layers, highlighting the long-term goal of supporting “infinite” context length, and “RoPE” refers to the rotary position embeddings employed in most layers.



Q & A